TM Forum Specification

Resource Role Management API User Guide

TMF768

Maturity Level: General Available (GA)	**Team Approved Date: 14-Nov-2024**
Release Status: Preview	**Approval Status: Team Approved**
Version 5.0.0	**IPR Mode: RAND**

© TM Forum 2024. All Rights Reserved.

NOTICE

Copyright © TM Forum 2024. All Rights Reserved.

This document and translations of it may be copied and furnished to others, and derivative works that comment on or otherwise explain it or assist in its implementation may be prepared, copied, published, and distributed, in whole or in part, without restriction of any kind, provided that the above copyright notice and this section are included on all such copies and derivative works. However, this document itself may not be modified in any way, including by removing the copyright notice or references to TM FORUM, except as needed for the purpose of developing any document or deliverable produced by a TM FORUM Collaboration Project Team (in which case the rules applicable to copyrights, as set forth in the TM FORUM IPR Policy, must be followed) or as required to translate it into languages other than English.

The limited permissions granted above are perpetual and will not be revoked by TM FORUM or its successors or assigns.

This document and the information contained herein is provided on an "AS IS" basis and TM FORUM DISCLAIMS ALL WARRANTIES, EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO ANY WARRANTY THAT THE USE OF THE INFORMATION HEREIN WILL NOT INFRINGE ANY OWNERSHIP RIGHTS OR ANY IMPLIED WARRANTIES OF MERCHANTABILITY OR FITNESS FOR A PARTICULAR PURPOSE.

Direct inquiries to the TM Forum office:

181 New Road, Suite 304
Parsippany, NJ 07054, USA
Tel No. +1 862 227 1648
TM Forum Web Page: www.tmforum.org

Table of Contents

NOTICE . . . i
Introduction . . . 1
Sample Use Cases . . . 6
Support of polymorphism and extension patterns . . . 7
RESOURCE MODEL . . . 8
Managed Entity and Task Resource Models . . . 8
ResourceRoleSpecification resource . . . 8
ResourceRole resource . . . 12
Notification Resource Models . . . 16
ResourceRoleSpecification . . . 17
ResourceRole . . . 22
API OPERATIONS . . . 27
Operations on ResourceRoleSpecification . . . 27
Retrieves a ResourceRoleSpecification by ID . . . 27
List or find ResourceRoleSpecification objects . . . 29
Creates a ResourceRoleSpecification . . . 31
Updates partially a ResourceRoleSpecification . . . 34
Deletes a ResourceRoleSpecification . . . 41
Operations on ResourceRole . . . 42
Retrieves a ResourceRole by ID . . . 42
List or find ResourceRole objects . . . 43
Creates a ResourceRole . . . 45
Updates partially a ResourceRole . . . 47
Deletes a ResourceRole . . . 53
API NOTIFICATIONS . . . 55
Register listener . . . 55
Unregister listener . . . 56
Publish Event to listener . . . 56
Acknowledgements . . . 58
Version History . . . 58
Release History . . . 58
Contributors to Document . . . 58

© TM Forum 2024. All Rights Reserved.

Introduction

The scope of this document is to define the Resource Role Management API.

A resource role, in a high-level context, is a designation assigned to a resource that defines its function, responsibilities, or usage in a particular context.

The Resource Role Management API is designed to facilitate the management of resource role instances and specifications. These specifications delineate the shared characteristics and behaviors inherent to each type of resource entity. Furthermore, this API provides comprehensive management capabilities for various facets of a resource role, ensuring efficient and effective control over the system's resources.

The data model for the Resource Role Management API is constructed based on the aggregated business entity model from the Information Framework (SID). This is followed by a concise explanation of the concepts associated with the roles of Resources. This approach ensures a comprehensive understanding of the API profile document.

Resource Role A Resource Role is a concept that delineates the diverse roles associated with Resources, encompassing both physical and logical aspects, within the model. This definition provides a structured understanding of the interactions and functionalities of the Resources in the system.

© TM Forum 2024. All Rights Reserved.

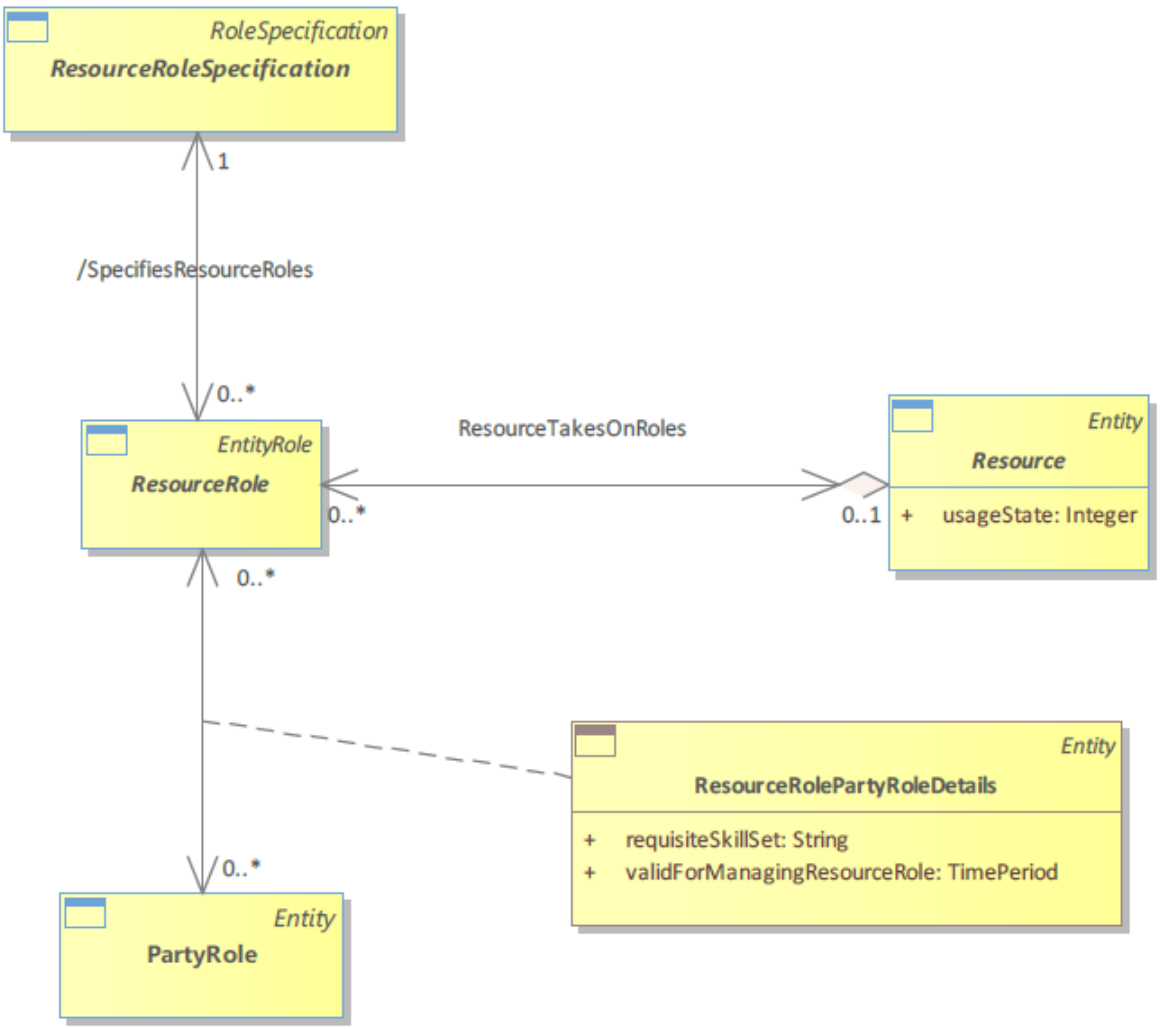


Figure 1. Resource role, a starting point (SID v23.0 Resource Domain Resource ABE)

Resource Specification might be described by Resource Roles that can be played by corresponding Resources.

Resource Role is sub-classed according to Resource Specification sub-classes.

© TM Forum 2024. All Rights Reserved.

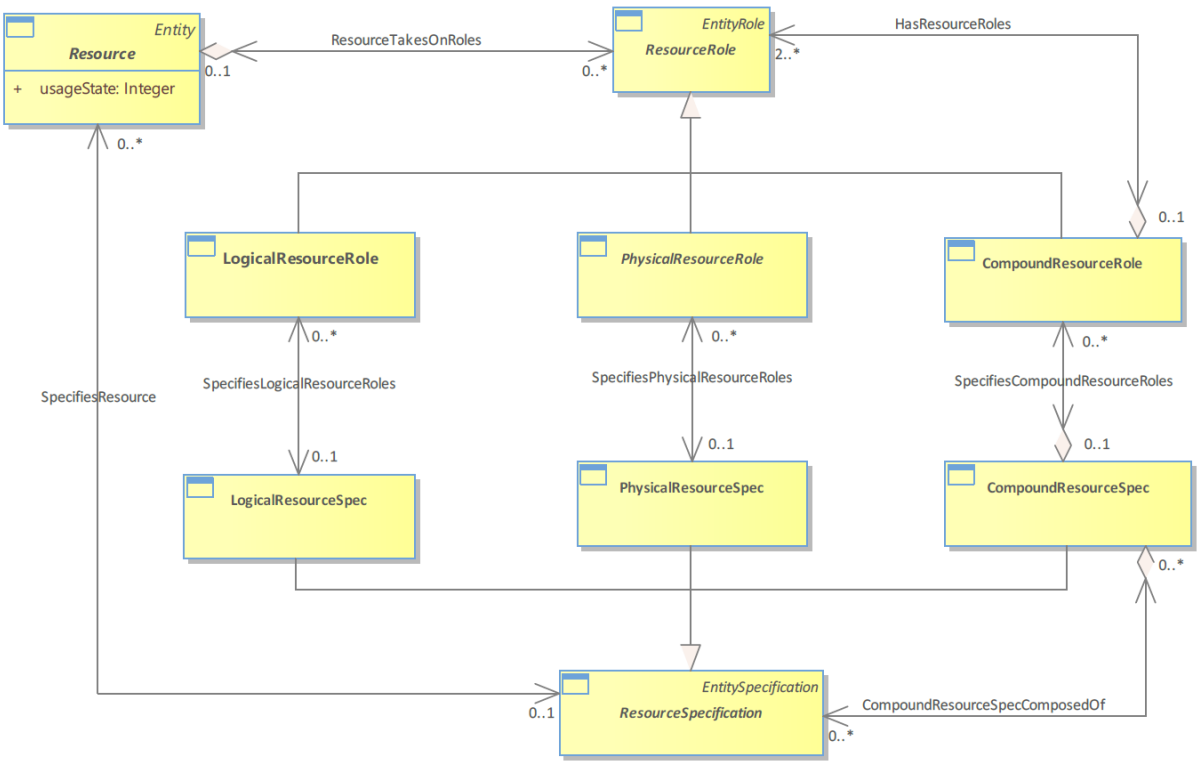


Figure 2. Extending roles (SID v23.0 Resource Domain Resource Specification ABE)

The Resource Role is designed with extensibility in mind, allowing for interactions between Resources and other business entities, such as Party Roles.

For instance, the following diagram illustrates the association of Party Roles with distinct Resource Roles, such as device roles.

This concept can be further elucidated with a practical business scenario: In a typical Provider network, the core and the edge utilize different devices due to their distinct roles. The core, serving a different purpose than the edge, necessitates the assignment of management personnel with varying skill sets for the core and edge devices. This differentiation in management is a direct result of the unique roles and requirements of the core and edge devices within the network.

© TM Forum 2024. All Rights Reserved.

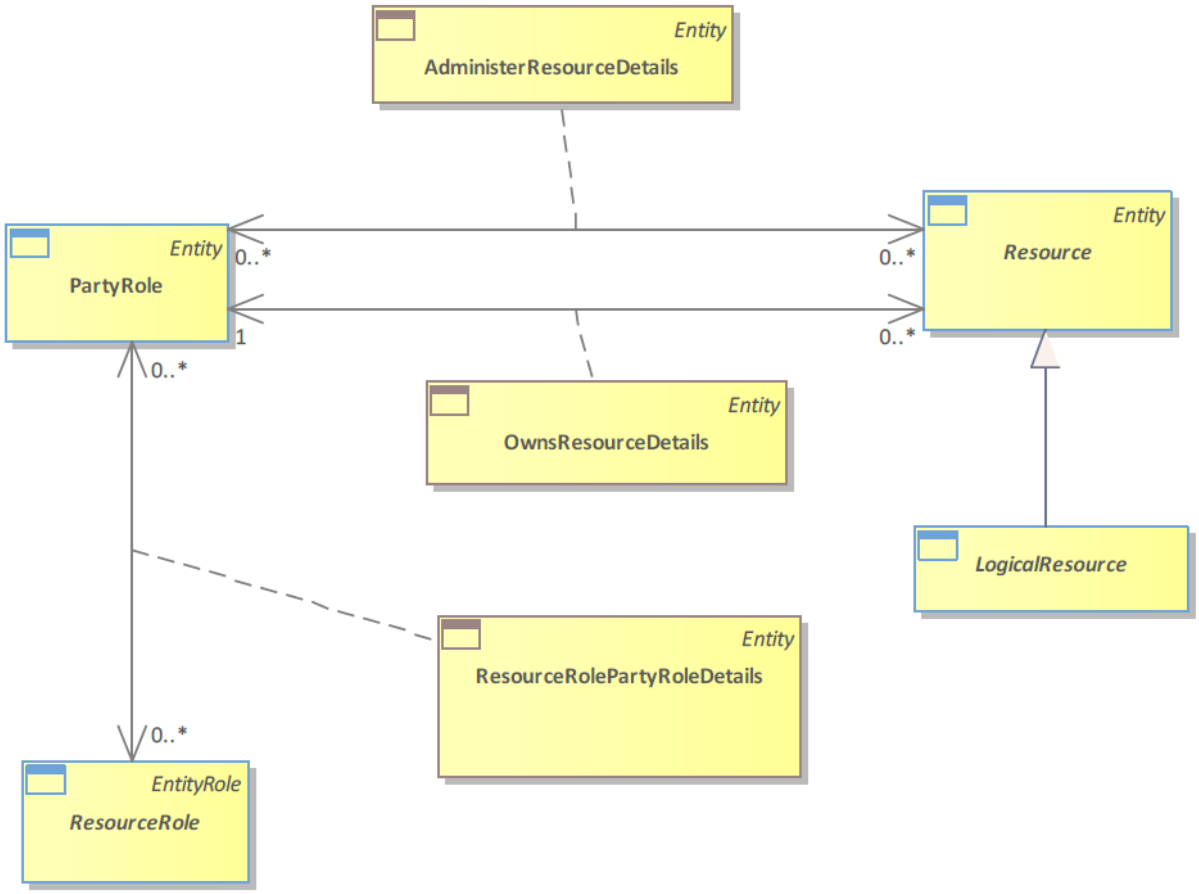


Figure 3. Managing resource roles using party roles (SID v23.0 Resource Domain Resource ABE)

Digital Identity Global Party and Resource Domain Overview Digital Identity API (TMF720) plays an important role in the global party and resource domains relating to User Role Permission API (TMF672), Party API (TMF632), Party Role API (TMF669), Federated Identity API (TMF691), Resource Inventory API (TMF639) and Resource Role API (TMF768), to managing and controlling access to information and resources within an organization or network.

In summary, these TM Forum Open APIs are related to managing parties, resources and their associated roles and digital identities within a network, as well as controlling access to information and resources through user roles and permissions. They provide a standardized way for organizations to manage these processes and enable integration with other systems.

Following diagram illustrates the digital identity in a global party and resource domains overview, referencing related TM Forum Open APIs and data model entities.

© TM Forum 2024. All Rights Reserved.

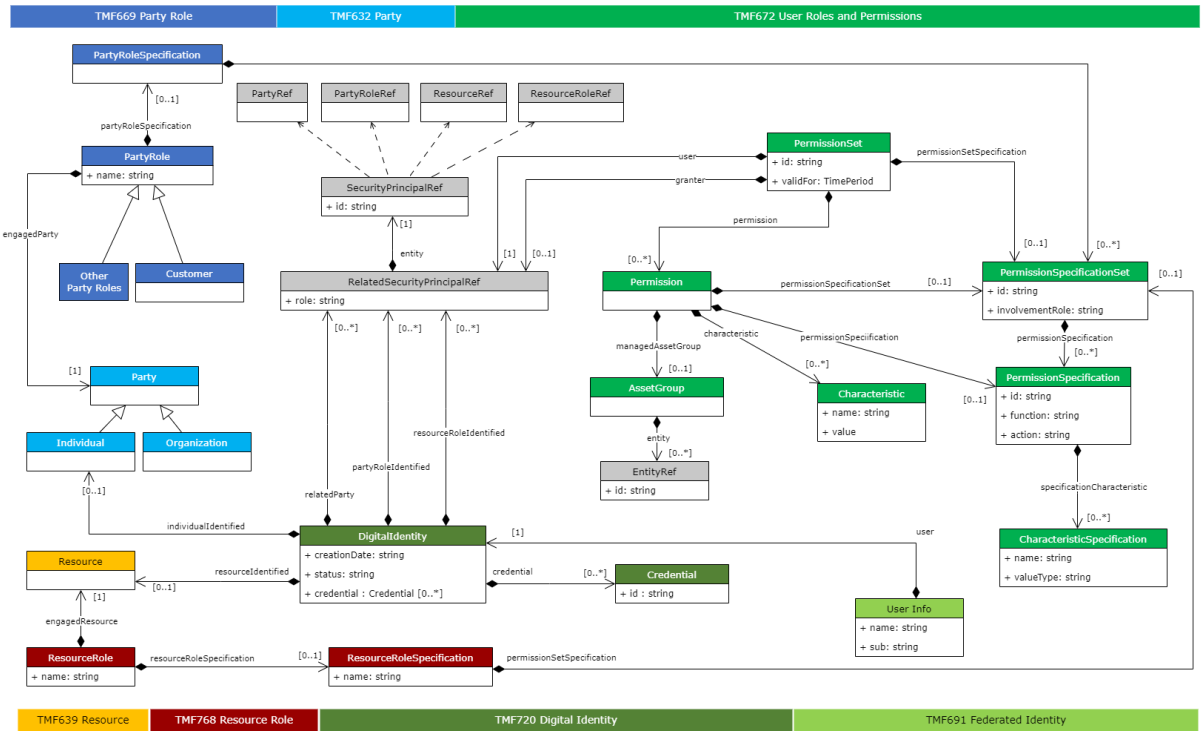


Figure 4. Digital identity global party and resource domain overview

The Resource Role Management API provides the capability of creating, updating, and deleting resource role specifications and resource roles.

The key functions are:

- Search resource role specification(s).
- Retrieve a resource role specification.
- Create a resource role specification.
- Update a resource role specification.
- Delete a resource role specification.
- Search resource role(s).
- Retrieve a resource role.
- Create a resource role.
- Update a resource role.
- Delete a resource role

© TM Forum 2024. All Rights Reserved.

Sample Use Cases

This section provides examples of how the most common business functions are implemented using the API standard.

The Resource Role Management API provides the ability to manage resource roles and its underlying specifications.

Some example use cases are:

- **UC1**: An Administration User of a Resource Catalog Management application creating a new resource role “Api Consumer” for a Product Catalog Management Web Application that needs to access protected resources such as TMF620 Product Catalog API.
- **UC2**: An Administration User of a Resource Catalog Management application creating a new resource role “Api Provider” for the TMF620 API to declare the full set of permissions exposed by the API.

As an example, at the end of both use cases the information created will be:

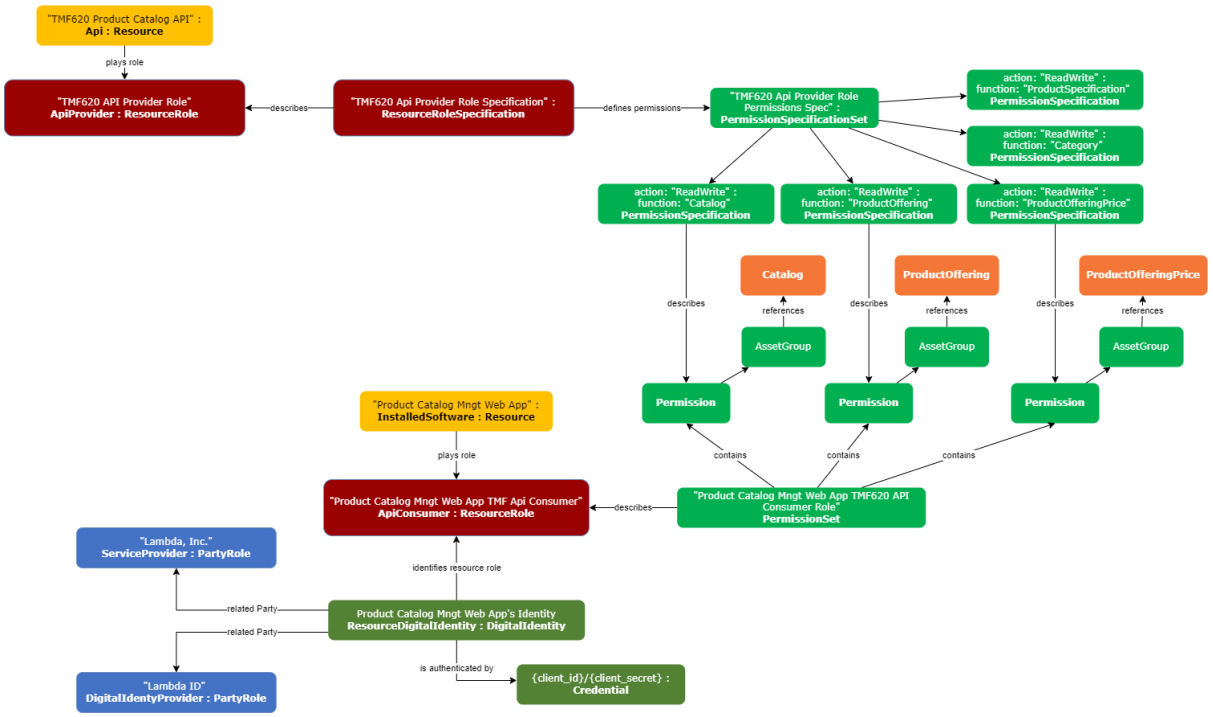


Figure 5. Product Catalog Management Web App Api Consumer example

© TM Forum 2024. All Rights Reserved.

Support of polymorphism and extension patterns

Support of polymorphic collections and types and schema based extension is provided by means of a list of generic meta-attributes that we describe below. Polymorphism in collections occurs when entities inherit from base entities, for instance a BillingAccount and SettlementAccount inheriting properties from the abstract Account entity.

Generic support of polymorphism and pattern extensions is described in the TMF API Guidelines, Part 2 (TMF630).

The @type attribute provides a way to represent the actual class type of an entity. For example, within a list of Account instances some may be instances of BillingAccount where other could be instances of SettlementAccount. The @type gives this information. All resources and sub-resources of this API have a @type attributes that can be provided when this is useful.

The @referredType can be used within reference entities (like for instance an AccountRef object) to explicitly denote the actual entity type of the referred class. Notice that in reference entities the @type, when used, denotes the class type of the reference itself, such as BillingAccountRef or SettlementAccountRef, and not the class type of the referred object. However since reference classes are rarely sub-classed, @type is generally not useful in reference objects.

The @schemaLocation property can be used in resources to allow specifying user-defined properties of an Entity or to specify the expected characteristics of an entity.

The @baseType attribute gives a way to provide explicitly the base of class of a given resource that has been extended.

© TM Forum 2024. All Rights Reserved.

RESOURCE MODEL

Managed Entity and Task Resource Models

ResourceRoleSpecification resource

Resource role specification.

Resource model

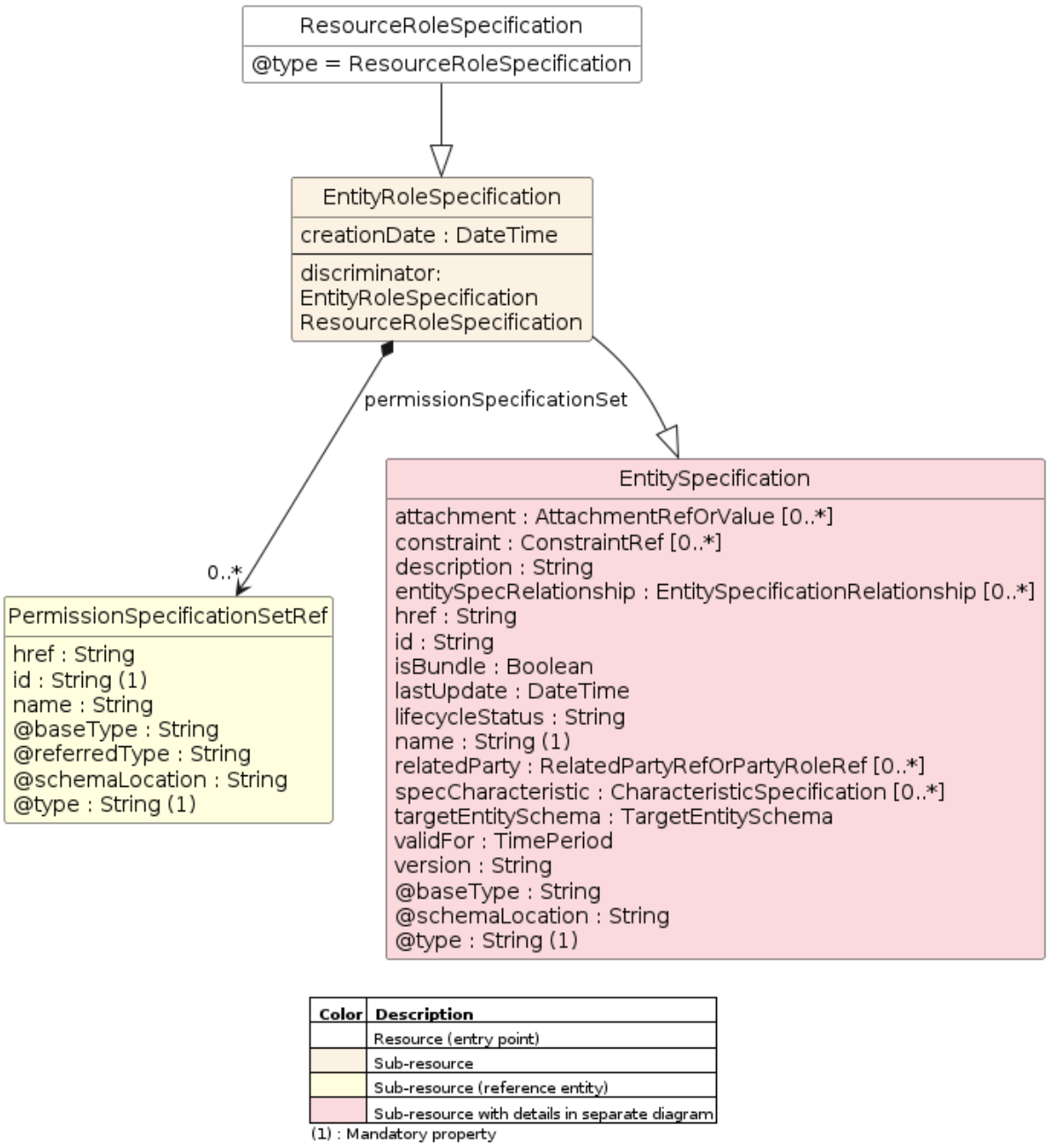


Figure 1 - ResourceRoleSpecification

© TM Forum 2024. All Rights Reserved.

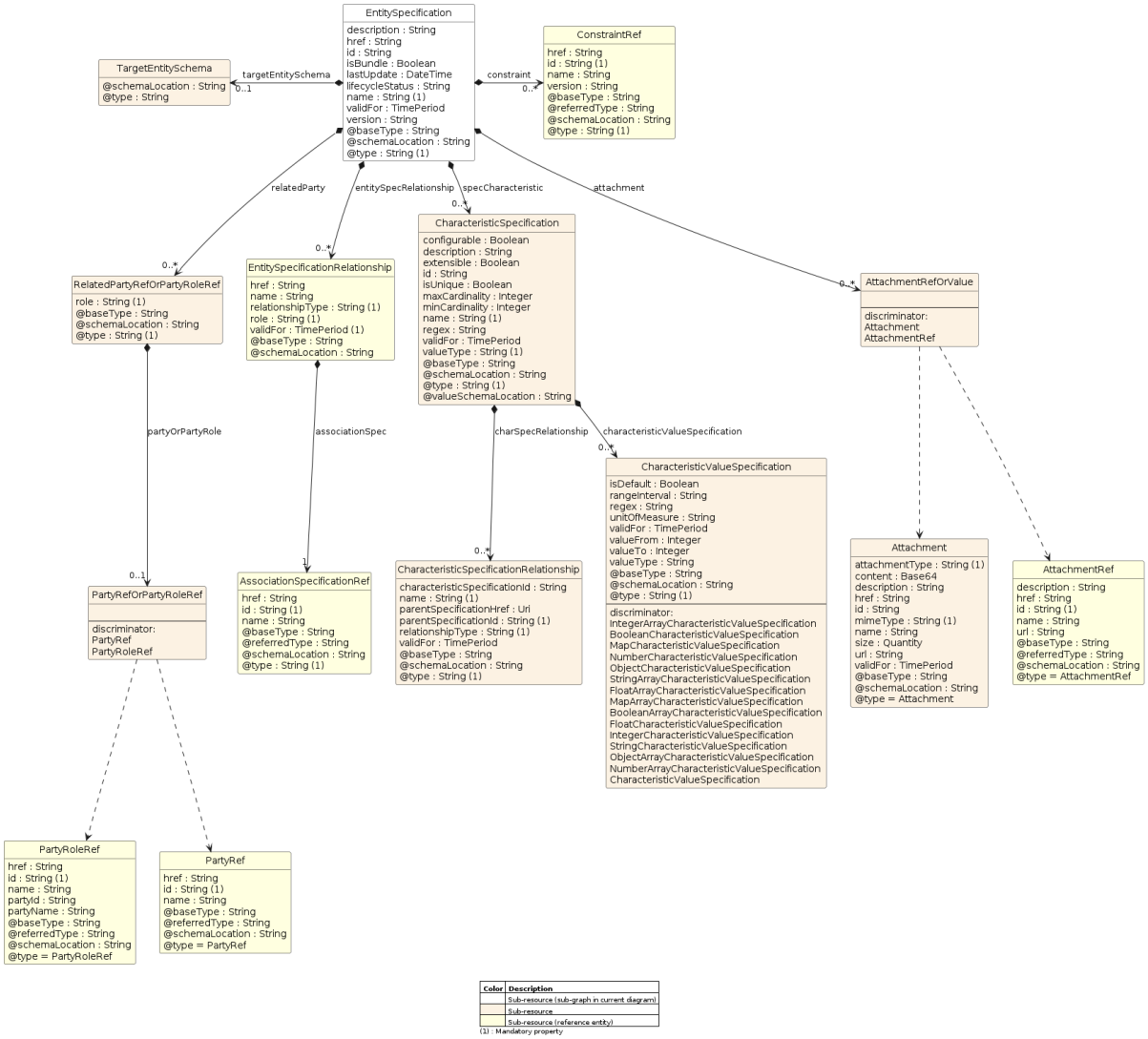


Figure 2 - EntitySpecification

Field descriptions

ResourceRoleSpecification fields

attachment	An AttachmentRefOrValue. The polymorphic attributes @type, @schemaLocation & @referredType are related to the Attachment entity and not the AttachmentRefOrValue class itself.
constraint	A ConstraintRef. Constraint reference. The Constraint resource represents a policy/rule applied to an entity or entity spec.
creationDate	A DateTime. Date and time of creation of the entity role specification.
description	A String. Description of the specification.
entitySpecRelationship	An EntitySpecificationRelationship. A migration, substitution, dependency or exclusivity relationship between/among entity specifications.
href	A String. Hyperlink reference.

© TM Forum 2024. All Rights Reserved.

id	A String. Unique identifier.
isBundle	A Boolean. IsBundle determines whether specification represents a single specification (false), or a bundle of specifications (true).
lastUpdate	A DateTime. Date and time of the last update of the specification.
lifecycleStatus	A String. Used to indicate the current lifecycle status of this catalog item.
name	A String. Name given to the specification.
permissionSpecificationSet	A PermissionSpecificationSetRef. A PermissionSpecificationSet defines a set of permissions that can be attached to a party role specification. The resulting permissions will then apply to all users that adopt this party role.
relatedParty	A RelatedPartyRefOrPartyRoleRef. RelatedParty reference. A related party defines party or party role or its reference, linked to a specific entity.
specCharacteristic	A CharacteristicSpecification. This class defines a characteristic specification.
targetEntitySchema	A TargetEntitySchema. The reference object to the schema and type of target entity which is described by a specification.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
version	A String. Specification version.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

Json representation sample(s)

We provide below a JSON representation as example of the 'ResourceRoleSpecification' resource object.


```
{
   "@type": "ResourceRoleSpecification",
   "href": "https://serverRoot/tmf-
api/resourceRoleManagement/v5/resourceRoleSpecification/2391ee4acbc142b1a4b74b4d4ab3
8152",
   "id": "2391ee4acbc142b1a4b74b4d4ab38152",
   "name": "TMF620 API provider role specification",
   "lastUpdate": "2024-02-20T17:00:00Z",
   "lifecycleStatus": "active",
   "validFor": {
      "startDateTime": "2024-02-20T17:00:00Z",
      "endDateTime": "2099-01-11T00:00:00Z"
   },
   "version": "1.0",
   "specCharacteristic": [
```


© TM Forum 2024. All Rights Reserved.

```
      {
         "@type": "CharacteristicSpecification",
         "name": "identifier",
         "valueType": "string",
         "description": "Unique identifier for the API. This value will be used as
the audience parameter on authorization calls.",
         "configurable": true,
         "minCardinality": 1,
         "maxCardinality": 1
      },
      {
         "@type": "CharacteristicSpecification",
         "name": "maximumAccessTokenLifetime",
         "valueType": "integer",
         "description": "Time until an access token issued for this API will
expire.",
         "configurable": true,
         "minCardinality": 1,
         "maxCardinality": 1
      },
      {
         "@type": "CharacteristicSpecification",
         "name": "jwtTokenSigningAlgorithm",
         "valueType": "string",
         "description": "Algorithm used to sign access tokens issued for this API.",
         "configurable": true,
         "minCardinality": 1,
         "maxCardinality": 1,
         "characteristicValueSpecification": [
            {
               "@type": "StringCharacteristicValueSpecification",
               "@baseType": "characteristicValueSpecification",
               "isDefault": true,
               "valueType": "string",
               "value": "RS256 (RSA Signature with SHA-256)"
            },
            {
               "@type": "StringCharacteristicValueSpecification",
               "@baseType": "characteristicValueSpecification",
               "valueType": "string",
               "value": "HS256 (HMAC with SHA-256)"
            },
            {
               "@type": "StringCharacteristicValueSpecification",
               "@baseType": "characteristicValueSpecification",
               "valueType": "string",
               "value": "PS256 (RSA Signature with SHA-256)"
            }
         ]
      }
   ],
   "permissionSpecificationSet": [
      {
         "href": "https://serverRoot/tmf-
api/rolesAndPermissionsManagement/v5/permissionSpecificationSet/193c8d6887394ca5b502
90685db512ca",
         "id": "193c8d6887394ca5b50290685db512ca",
         "@type": "PermissionSpecificationSetRef"
      }
   ],
   "creationDate": "2024-02-20T17:00:00Z"
}
```


© TM Forum 2024. All Rights Reserved.

ResourceRole resource

The part played by a resource in a given context.

Resource model

© TM Forum 2024. All Rights Reserved.

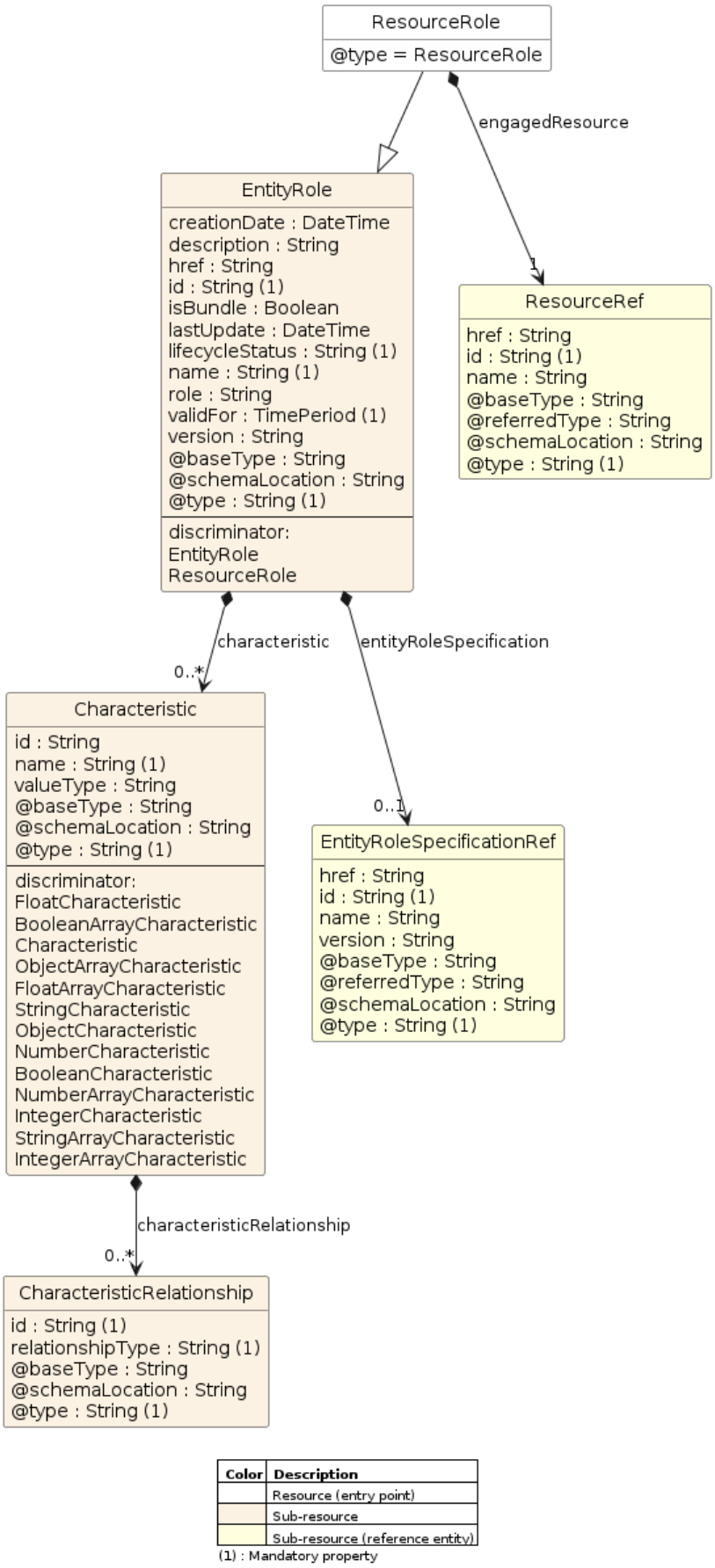


Figure 3 - ResourceRole

© TM Forum 2024. All Rights Reserved.

Field descriptions

ResourceRole fields

characteristic	A Characteristic. Describes a given characteristic of an object or entity through a name/value pair. This is an abstract base class, the actual value is in one of the strongly-typed subclasses : StringCharacteristic, ObjectCharacteristic, FloatCharacteristic, BooleanCharacteristic, NumberCharacteristic, IntegerCharacteristic, StringArrayCharacteristic, ObjectArrayCharacteristic, BooleanArrayCharacteristic, NumberArrayCharacteristic, IntegerArrayCharacteristic… Characteristic can be instanciated as * BooleanArrayCharacteristic * BooleanCharacteristic * FloatArrayCharacteristic * FloatCharacteristic * IntegerArrayCharacteristic * IntegerCharacteristic * NumberArrayCharacteristic * NumberCharacteristic * ObjectArrayCharacteristic * ObjectCharacteristic * StringArrayCharacteristic * StringCharacteristic
creationDate	A DateTime. Date and time of creation of the entity role.
description	A String. Description of the specification.
engagedResource	A ResourceRef. Resource reference, for when Resource is used by other entities.
entityRoleSpecification	An EntityRoleSpecificationRef. Reference to an EntitySpecification object.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
isBundle	A Boolean. IsBundle determines whether specification represents a single specification (false), or a bundle of specifications (true).
lastUpdate	A DateTime. Date and time of the last update of the specification.
lifecycleStatus	A String. Used to indicate the current lifecycle status of this catalog item.
name	A String. Name given to the specification.
role	A String. Role played by the engagedEntity in this context. As role is defined by entityRoleSpecification, this role attribute can be used to precise the role defined by entityRoleSpecification, or it can be used to define the role in case there is no entityRoleSpecification.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.

© TM Forum 2024. All Rights Reserved.

version	A String. Specification version.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ResourceRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

Json representation sample(s)

We provide below a JSON representation as example of the 'ResourceRole' resource object.


```
{
   "@type": "ResourceRole",
   "href": "https://serverRoot/tmf-
api/resourceRoleManagement/v5/resourceRole/2391ee4acbc142b1a4b74b4d4ab38152",
   "id": "2391ee4acbc142b1a4b74b4d4ab38152",
   "name": "TMF620 API provider resource role",
   "lastUpdate": "2024-02-20T17:00:00Z",
   "lifecycleStatus": "active",
   "validFor": {
      "startDateTime": "2024-02-20T17:00:00Z",
      "endDateTime": "2099-01-11T00:00:00Z"
   },
   "version": "1.0",
   "role": "Api Provider",
   "entityRoleSpecification": {
      "href": "https://serverRoot/tmf-
api/resourceRoleManagement/v5/resourceRoleSpecification/2391ee4acbc142b1a4b74b4d4ab3
8152",
      "id": "2391ee4acbc142b1a4b74b4d4ab38152",
      "@type": "EntityRoleSpecificationRef",
      "@referredType": "ResourceRoleSpecification"
   },
   "characteristic": [
      {
         "@type": "StringCharacteristic",
         "@baseType": "Characteristic",
         "name": "identitier",
```


© TM Forum 2024. All Rights Reserved.

```
            "valueType": "string",
            "value": "https://serverRoot/tmf-api/productCatalogManagement/v5"
         },
         {
            "@type": "IntegerCharacteristic",
            "@baseType": "Characteristic",
            "name": "maximumAccessTokenLifetime",
            "valueType": "integer",
            "value": 86400
         },
         {
            "@type": "StringCharacteristic",
            "@baseType": "Characteristic",
            "name": "jwtSigningAlgorithm",
            "valueType": "string",
            "value": "RS256 (RSA Signature with SHA-256)"
         }
      ],
      "creationDate": "2024-02-20T17:00:00Z",
      "engagedResource": {
         "href": "https://serverRoot/tmf-
api/resourceManagement/v5/resource/e3479404feb943d2af7af92ffe4ec2b0",
         "id": "e3479404feb943d2af7af92ffe4ec2b0",
         "@type": "ResourceRef",
         "@referredType": "Resource"
      }
   }
```


Notification Resource Models

8 notifications are defined for this API.

Notifications related to ResourceRoleSpecification:

- Create Event
- Delete Event
- Attribute Value Change Event
- Status Change Event

Notifications related to ResourceRole:

- Create Event
- Delete Event
- Attribute Value Change Event
- Status Change Event

The notification structure for all notifications in this API follow the pattern depicted by the figure below. A notification event resource (depicted by "SpecificEvent" placeholder) is a sub class of a generic Event structure containing at least an id of the event occurrence (eventId), an event timestamp (eventTime), and the name of the resource (eventType). This notification structure owns an event payload structure ("SpecificEventPayload" placeholder) linked to the resource concerned by the notification using the resource name as access field ("resourceName" placeholder).

© TM Forum 2024. All Rights Reserved.

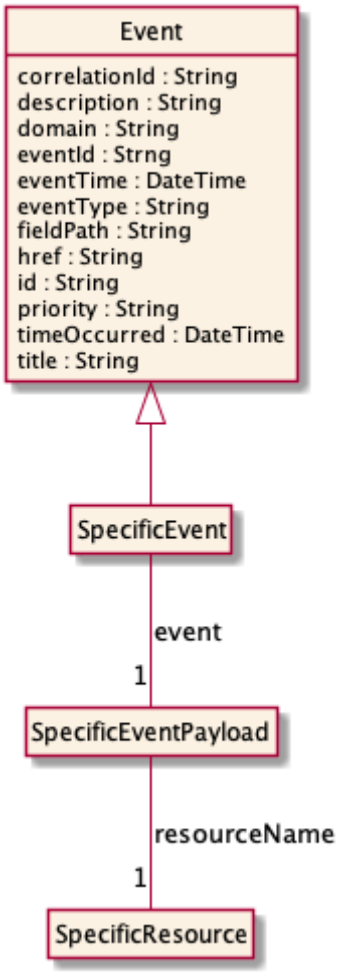


Figure 4 Notification Pattern

ResourceRoleSpecification

Create Event

Message example for ResourceRoleSpecificationCreate Event Notification

```
Content-Type: application/json

{
   "correlationId": "95003dd3-e325",
   "description": "ResourceRoleSpecificationCreateEvent illustration",
   "domain": "Commercial",
   "eventId": "4010-9b59-509a64cf85a8",
   "eventTime": "2022-08-25T12:18:12.171Z",
   "eventType": "ResourceRoleSpecificationCreateEvent",
   "priority": "1",
   "timeOccurred": "2022-08-25T12:18:06.252Z",
   "title": "ResourceRoleSpecificationCreateEvent",
   "event": {
      "resourceRoleSpecification": {
         "@type": "ResourceRoleSpecification",
         "href": "https://serverRoot/tmf-
api/resourceRoleManagement/v5/resourceRoleSpecification/2391ee4acbc142b1a4b74b4d4ab3
8152",
```

© TM Forum 2024. All Rights Reserved.

```
          "id": "2391ee4acbc142b1a4b74b4d4ab38152",
          "name": "TMF620 API provider role specification",
          "lastUpdate": "2024-02-20T17:00:00Z",
          "lifecycleStatus": "active",
          "validFor": {
             "startDateTime": "2024-02-20T17:00:00Z",
             "endDateTime": "2099-01-11T00:00:00Z"
          },
          "version": "1.0",
          "specCharacteristic": [
             {
                "@type": "CharacteristicSpecification",
                "name": "identifier",
                "valueType": "string",
                "description": "Unique identifier for the API. This value will be
used as the audience parameter on authorization calls.",
                "configurable": true,
                "minCardinality": 1,
                "maxCardinality": 1
             },
             {
                "@type": "CharacteristicSpecification",
                "name": "maximumAccessTokenLifetime",
                "valueType": "integer",
                "description": "Time until an access token issued for this API will
expire.",
                "configurable": true,
                "minCardinality": 1,
                "maxCardinality": 1
             },
             {
                "@type": "CharacteristicSpecification",
                "name": "jwtTokenSigningAlgorithm",
                "valueType": "string",
                "description": "Algorithm used to sign access tokens issued for this
API.",
                "configurable": true,
                "minCardinality": 1,
                "maxCardinality": 1,
                "characteristicValueSpecification": [
                   {
                      "@type": "StringCharacteristicValueSpecification",
                      "@baseType": "characteristicValueSpecification",
                      "isDefault": true,
                      "valueType": "string",
                      "value": "RS256 (RSA Signature with SHA-256)"
                   },
                   {
                      "@type": "StringCharacteristicValueSpecification",
                      "@baseType": "characteristicValueSpecification",
                      "valueType": "string",
                      "value": "HS256 (HMAC with SHA-256)"
                   },
                   {
                      "@type": "StringCharacteristicValueSpecification",
                      "@baseType": "characteristicValueSpecification",
                      "valueType": "string",
                      "value": "PS256 (RSA Signature with SHA-256)"
                   }
                ]
             }
          ],
          "permissionSpecificationSet": [
```


© TM Forum 2024. All Rights Reserved.

```
            {
               "href": "https://serverRoot/tmf-
api/rolesAndPermissionsManagement/v5/permissionSpecificationSet/193c8d6887394ca5b502
90685db512ca",
               "id": "193c8d6887394ca5b50290685db512ca",
               "@type": "PermissionSpecificationSetRef"
            }
         ],
         "creationDate": "2024-02-20T17:00:00Z"
      }
   },
   "reportingSystem": {
      "id": "759",
      "name": "APP-745",
      "@type": "ReportingResource",
      "@referredType": "LogicalResource"
   },
   "source": {
      "id": "705",
      "name": "APP-317",
      "@type": "ReportingResource",
      "@referredType": "LogicalResource"
   },
   "@baseType": "Event",
   "@type": "ResourceRoleSpecificationCreateEvent"
}
```


Delete Event

Message example for ResourceRoleSpecificationDelete Event Notification


```
Content-Type: application/json

{
   "correlationId": "90cfc73d-deb7",
   "description": "ResourceRoleSpecificationDeleteEvent illustration",
   "domain": "Commercial",
   "eventId": "47d6-9751-40e4f01440c9",
   "eventTime": "2022-08-25T12:18:12.202Z",
   "eventType": "ResourceRoleSpecificationDeleteEvent",
   "priority": "4",
   "timeOccurred": "2022-08-25T12:18:07.224Z",
   "title": "ResourceRoleSpecificationDeleteEvent",
   "event": {
      "resourceRoleSpecification": {
         "@type": "ResourceRoleSpecification",
         "href": "https://serverRoot/tmf-
api/resourceRoleManagement/v5/resourceRoleSpecification/2391ee4acbc142b1a4b74b4d4ab3
8152",
         "id": "2391ee4acbc142b1a4b74b4d4ab38152"
      }
   },
   "reportingSystem": {
      "id": "759",
      "name": "APP-745",
      "@type": "ReportingResource",
      "@referredType": "LogicalResource"
   },
   "source": {
      "id": "705",
```


© TM Forum 2024. All Rights Reserved.

```
      "name": "APP-317",
      "@type": "ReportingResource",
      "@referredType": "LogicalResource"
   },
   "@baseType": "Event",
   "@type": "ResourceRoleSpecificationDeleteEvent"
}
```

Attribute Value Change Event

Message example for ResourceRoleSpecificationAttributeValueChange Event Notification

```
Content-Type: application/json

{
   "correlationId": "333-fd6",
   "description": "ResourceRoleSpecificationAttributeValueChangeEvent illustration",
   "domain": "Commercial",
   "eventId": "569",
   "eventTime": "2021-09-27T07:43:59.059Z",
   "eventType": "ResourceRoleSpecificationAttributeValueChangeEvent",
   "priority": "1",
   "timeOccurred": "2021-09-27T07:43:59.059Z",
   "title": "ResourceRoleSpecificationAttributeValueChangeEvent",
   "event": {
      "resourceRoleSpecification": {
         "@type": "ResourceRoleSpecification",
         "href": "https://serverRoot/tmf-
api/resourceRoleManagement/v5/resourceRoleSpecification/2391ee4acbc142b1a4b74b4d4ab3
8152",
         "id": "2391ee4acbc142b1a4b74b4d4ab38152",
         "specCharacteristic": [
            {
               "@type": "CharacteristicSpecification",
               "name": "identifier",
               "valueType": "string",
               "description": "Unique identifier for the API. This value will be
used as the audience parameter on authorization calls.",
               "configurable": true,
               "minCardinality": 1,
               "maxCardinality": 1
            },
            {
               "@type": "CharacteristicSpecification",
               "name": "maximumAccessTokenLifetime",
               "valueType": "integer",
               "description": "Time until an access token issued for this API will
expire.",
               "configurable": true,
               "minCardinality": 1,
               "maxCardinality": 1
            },
            {
               "@type": "CharacteristicSpecification",
               "name": "jwtTokenSigningAlgorithm",
               "valueType": "string",
               "description": "Algorithm used to sign access tokens issued for this
API.",
               "configurable": true,
               "minCardinality": 1,
```

© TM Forum 2024. All Rights Reserved.

```
                "maxCardinality": 1,
                "characteristicValueSpecification": [
                   {
                      "@type": "StringCharacteristicValueSpecification",
                      "@baseType": "characteristicValueSpecification",
                      "isDefault": true,
                      "valueType": "string",
                      "value": "RS256 (RSA Signature with SHA-256)"
                   },
                   {
                      "@type": "StringCharacteristicValueSpecification",
                      "@baseType": "characteristicValueSpecification",
                      "valueType": "string",
                      "value": "HS256 (HMAC with SHA-256)"
                   },
                   {
                      "@type": "StringCharacteristicValueSpecification",
                      "@baseType": "characteristicValueSpecification",
                      "valueType": "string",
                      "value": "PS256 (RSA Signature with SHA-256)"
                   }
                ]
             }
          ],
          "permissionSpecificationSet": [
             {
                "href": "https://serverRoot/tmf-
api/rolesAndPermissionsManagement/v5/permissionSpecificationSet/193c8d6887394ca5b502
90685db512ca",
                "id": "193c8d6887394ca5b50290685db512ca",
                "@type": "PermissionSpecificationSetRef"
             }
          ]
       }
    },
    "reportingSystem": {
       "id": "123",
       "name": "CRM app",
       "@type": "ReportingResource",
       "@referredType": "LogicalResource"
    },
    "source": {
       "id": "123",
       "name": "CRM app",
       "@type": "ReportingResource",
       "@referredType": "LogicalResource"
    },
    "@baseType": "Event",
    "@type": "ResourceRoleSpecificationAttributeValueChangeEvent"
}
```


Status Change Event

Message example for ResourceRoleSpecificationStatusChange Event Notification


```
Content-Type: application/json

{
   "correlationId": "0f874cb9-c70d",
   "description": "ResourceRoleSpecificationStatusChangeEvent illustration",
```


© TM Forum 2024. All Rights Reserved.

```
    "domain": "Commercial",
    "eventId": "41cf-ad81-d65ea1c2840c",
    "eventTime": "2022-08-25T12:18:12.191Z",
    "eventType": "ResourceRoleSpecificationStatusChangeEvent",
    "priority": "5",
    "timeOccurred": "2022-08-25T12:18:08.202Z",
    "title": "ResourceRoleSpecificationStatusChangeEvent",
    "event": {
       "resourceRoleSpecification": {
          "@type": "ResourceRoleSpecification",
          "href": "https://serverRoot/tmf-
api/resourceRoleManagement/v5/resourceRoleSpecification/2391ee4acbc142b1a4b74b4d4ab3
8152",
          "id": "2391ee4acbc142b1a4b74b4d4ab38152",
          "lifecycleStatus": "active"
       }
    },
    "reportingSystem": {
       "id": "759",
       "name": "APP-745",
       "@type": "ReportingResource",
       "@referredType": "LogicalResource"
    },
    "source": {
       "id": "705",
       "name": "APP-317",
       "@type": "ReportingResource",
       "@referredType": "LogicalResource"
    },
    "@baseType": "Event",
    "@type": "ResourceRoleSpecificationStatusChangeEvent"
}
```


ResourceRole

Create Event

Message example for ResourceRoleCreate Event Notification


```
Content-Type: application/json

{
    "correlationId": "95003dd3-e325",
    "description": "ResourceRoleCreateEvent illustration",
    "domain": "Commercial",
    "eventId": "4010-9b59-509a64cf85a8",
    "eventTime": "2022-08-25T12:18:12.171Z",
    "eventType": "ResourceRoleCreateEvent",
    "priority": "1",
    "timeOccurred": "2022-08-25T12:18:06.252Z",
    "title": "ResourceRoleCreateEvent",
    "event": {
       "resourceRole": {
          "@type": "ResourceRole",
          "href": "https://serverRoot/tmf-
api/resourceRoleManagement/v5/resourceRole/2391ee4acbc142b1a4b74b4d4ab38152",
          "id": "2391ee4acbc142b1a4b74b4d4ab38152",
          "name": "TMF620 API provider resource role",
          "lastUpdate": "2024-02-20T17:00:00Z",
```


© TM Forum 2024. All Rights Reserved.

```
         "lifecycleStatus": "active",
         "validFor": {
            "startDateTime": "2024-02-20T17:00:00Z",
            "endDateTime": "2099-01-11T00:00:00Z"
         },
         "version": "1.0",
         "role": "Api Provider",
         "entityRoleSpecification": {
            "href": "https://serverRoot/tmf-
api/resourceRoleManagement/v5/resourceRoleSpecification/2391ee4acbc142b1a4b74b4d4ab3
8152",
            "id": "2391ee4acbc142b1a4b74b4d4ab38152",
            "@type": "EntityRoleSpecificationRef",
            "@referredType": "ResourceRoleSpecification"
         },
         "characteristic": [
            {
               "@type": "StringCharacteristic",
               "@baseType": "Characteristic",
               "name": "identitier",
               "valueType": "string",
               "value": "https://serverRoot/tmf-api/productCatalogManagement/v5"
            },
            {
               "@type": "IntegerCharacteristic",
               "@baseType": "Characteristic",
               "name": "maximumAccessTokenLifetime",
               "valueType": "integer",
               "value": 86400
            },
            {
               "@type": "StringCharacteristic",
               "@baseType": "Characteristic",
               "name": "jwtSigningAlgorithm",
               "valueType": "string",
               "value": "RS256 (RSA Signature with SHA-256)"
            }
         ],
         "creationDate": "2024-02-20T17:00:00Z",
         "engagedResource": {
            "href": "https://serverRoot/tmf-
api/resourceManagement/v5/resource/e3479404feb943d2af7af92ffe4ec2b0",
            "id": "e3479404feb943d2af7af92ffe4ec2b0",
            "@type": "ResourceRef",
            "@referredType": "Resource"
         }
      }
   },
   "reportingSystem": {
      "id": "759",
      "name": "APP-745",
      "@type": "ReportingResource",
      "@referredType": "LogicalResource"
   },
   "source": {
      "id": "705",
      "name": "APP-317",
      "@type": "ReportingResource",
      "@referredType": "LogicalResource"
   },
   "@baseType": "Event",
   "@type": "ResourceRoleCreateEvent"
```


© TM Forum 2024. All Rights Reserved.

```
}
```

Delete Event

Message example for ResourceRoleDelete Event Notification

```
Content-Type: application/json

{
   "correlationId": "90cfc73d-deb7",
   "description": "ResourceRoleDeleteEvent illustration",
   "domain": "Commercial",
   "eventId": "47d6-9751-40e4f01440c9",
   "eventTime": "2022-08-25T12:18:12.202Z",
   "eventType": "ResourceRoleDeleteEvent",
   "priority": "4",
   "timeOccurred": "2022-08-25T12:18:07.224Z",
   "title": "ResourceRoleDeleteEvent",
   "event": {
      "resourceRole": {
         "@type": "ResourceRole",
         "href": "https://serverRoot/tmf-
api/resourceRoleManagement/v5/resourceRole/2391ee4acbc142b1a4b74b4d4ab38152",
         "id": "2391ee4acbc142b1a4b74b4d4ab38152"
      }
   },
   "reportingSystem": {
      "id": "759",
      "name": "APP-745",
      "@type": "ReportingResource",
      "@referredType": "LogicalResource"
   },
   "source": {
      "id": "705",
      "name": "APP-317",
      "@type": "ReportingResource",
      "@referredType": "LogicalResource"
   },
   "@baseType": "Event",
   "@type": "ResourceRoleDeleteEvent"
}
```

Attribute Value Change Event

Message example for ResourceRoleAttributeValueChange Event Notification

```
Content-Type: application/json

{
   "correlationId": "333-fd6",
   "description": "ResourceRoleAttributeValueChangeEvent illustration",
   "domain": "Commercial",
   "eventId": "569",
   "eventTime": "2021-09-27T07:43:59.059Z",
   "eventType": "ResourceRoleAttributeValueChangeEvent",
   "priority": "1",
   "timeOccurred": "2021-09-27T07:43:59.059Z",
```

© TM Forum 2024. All Rights Reserved.

```
    "title": "ResourceRoleAttributeValueChangeEvent",
    "event": {
       "resourceRole": {
          "@type": "ResourceRole",
          "href": "https://serverRoot/tmf-
api/resourceRoleManagement/v5/resourceRole/2391ee4acbc142b1a4b74b4d4ab38152",
          "id": "2391ee4acbc142b1a4b74b4d4ab38152",
          "engagedResource": {
             "href": "https://serverRoot/tmf-
api/resourceManagement/v5/resource/e3479404feb943d2af7af92ffe4ec2b0",
             "id": "e3479404feb943d2af7af92ffe4ec2b0",
             "@type": "ResourceRef"
          }
       }
    },
    "reportingSystem": {
       "id": "123",
       "name": "CRM app",
       "@type": "ReportingResource",
       "@referredType": "LogicalResource"
    },
    "source": {
       "id": "123",
       "name": "CRM app",
       "@type": "ReportingResource",
       "@referredType": "LogicalResource"
    },
    "@baseType": "Event",
    "@type": "ResourceRoleAttributeValueChangeEvent"
}
```


Status Change Event

Message example for ResourceRoleStatusChange Event Notification


```
Content-Type: application/json

{
    "correlationId": "0f874cb9-c70d",
    "description": "ResourceRoleStatusChangeEvent illustration",
    "domain": "Commercial",
    "eventId": "41cf-ad81-d65ea1c2840c",
    "eventTime": "2022-08-25T12:18:12.191Z",
    "eventType": "ResourceRoleStatusChangeEvent",
    "priority": "5",
    "timeOccurred": "2022-08-25T12:18:08.202Z",
    "title": "ResourceRoleStatusChangeEvent",
    "event": {
       "resourceRole": {
          "@type": "ResourceRole",
          "href": "https://serverRoot/tmf-
api/resourceRoleManagement/v5/resourceRole/2391ee4acbc142b1a4b74b4d4ab38152",
          "id": "2391ee4acbc142b1a4b74b4d4ab38152",
          "lifecycleStatus": "active"
       }
    },
    "reportingSystem": {
       "id": "759",
       "name": "APP-745",
       "@type": "ReportingResource",
```


© TM Forum 2024. All Rights Reserved.

```
        "@referredType": "LogicalResource"
    },
    "source": {
        "id": "705",
        "name": "APP-317",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "@baseType": "Event",
    "@type": "ResourceRoleStatusChangeEvent"
}
```


© TM Forum 2024. All Rights Reserved.

API OPERATIONS

Remember the following Uniform Contract:

Operation on Entities	Uniform API Operation	Description
Query Entities	GET Resource	GET must be used to retrieve a representation of a resource.
Create Entity	POST Resource	POST must be used to create a new resource
Partial Update of an Entity	PATCH Resource	PATCH must be used to partially update a resource
Remove an Entity	DELETE Resource	DELETE must be used to remove a resource
Execute an Action on an Entity	POST on TASK Resource	POST must be used to execute Task Resources
Other Request Methods	POST on TASK Resource	GET and POST must not be used to tunnel other request methods.

Filtering and attribute selection rules are described in the TMF REST Design Guidelines.

Notifications are also described in a subsequent section.

Operations on ResourceRoleSpecification

Retrieves a ResourceRoleSpecification by ID

`GET /resourceRoleSpecification/{id}?fields=…`

Description

This operation retrieves a ResourceRoleSpecification entity. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

Usage samples

Retrieve Resource Role Specification example.

Request

```
GET /resourceRoleSpecification/2391ee4acbc142b1a4b74b4d4ab38152
Content-Type: application/json
```

Response

```
200
```

© TM Forum 2024. All Rights Reserved.

```
{
   "@type": "ResourceRoleSpecification",
   "href": "https://serverRoot/tmf-
api/resourceRoleManagement/v5/resourceRoleSpecification/2391ee4acbc142b1a4b74b4d4ab3
8152",
   "id": "2391ee4acbc142b1a4b74b4d4ab38152",
   "name": "TMF620 API provider role specification",
   "lastUpdate": "2024-02-20T17:00:00Z",
   "lifecycleStatus": "active",
   "validFor": {
      "startDateTime": "2024-02-20T17:00:00Z",
      "endDateTime": "2099-01-11T00:00:00Z"
   },
   "version": "1.0",
   "specCharacteristic": [
      {
         "@type": "CharacteristicSpecification",
         "name": "identifier",
         "valueType": "string",
         "description": "Unique identifier for the API. This value will be used as
the audience parameter on authorization calls.",
         "configurable": true,
         "minCardinality": 1,
         "maxCardinality": 1
      },
      {
         "@type": "CharacteristicSpecification",
         "name": "maximumAccessTokenLifetime",
         "valueType": "integer",
         "description": "Time until an access token issued for this API will
expire.",
         "configurable": true,
         "minCardinality": 1,
         "maxCardinality": 1
      },
      {
         "@type": "CharacteristicSpecification",
         "name": "jwtTokenSigningAlgorithm",
         "valueType": "string",
         "description": "Algorithm used to sign access tokens issued for this API.",
         "configurable": true,
         "minCardinality": 1,
         "maxCardinality": 1,
         "characteristicValueSpecification": [
            {
               "@type": "StringCharacteristicValueSpecification",
               "@baseType": "characteristicValueSpecification",
               "isDefault": true,
               "valueType": "string",
               "value": "RS256 (RSA Signature with SHA-256)"
            },
            {
               "@type": "StringCharacteristicValueSpecification",
               "@baseType": "characteristicValueSpecification",
               "valueType": "string",
               "value": "HS256 (HMAC with SHA-256)"
            },
            {
               "@type": "StringCharacteristicValueSpecification",
               "@baseType": "characteristicValueSpecification",
               "valueType": "string",
               "value": "PS256 (RSA Signature with SHA-256)"
            }
```


© TM Forum 2024. All Rights Reserved.

```
         ]
      }
   ],
   "permissionSpecificationSet": [
      {
         "href": "https://serverRoot/tmf-
api/rolesAndPermissionsManagement/v5/permissionSpecificationSet/193c8d6887394ca5b502
90685db512ca",
         "id": "193c8d6887394ca5b50290685db512ca",
         "@type": "PermissionSpecificationSetRef"
      }
   ],
   "creationDate": "2024-02-20T17:00:00Z"
}
```

List or find ResourceRoleSpecification objects

`GET /resourceRoleSpecification?fields=…`

Description

This operation list ResourceRoleSpecification entities. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

Usage samples

Retrieve Resource Role Specifications example.

Request

```
GET /resourceRoleSpecification
Content-Type: application/json
```

Response

```
200

[
   {
      "@type": "ResourceRoleSpecification",
      "href": "https://serverRoot/tmf-
api/resourceRoleManagement/v5/resourceRoleSpecification/2391ee4acbc142b1a4b74b4d4ab3
8152",
      "id": "2391ee4acbc142b1a4b74b4d4ab38152",
      "name": "TMF620 API provider role specification",
      "lastUpdate": "2024-02-20T17:00:00Z",
      "lifecycleStatus": "active",
      "validFor": {
         "startDateTime": "2024-02-20T17:00:00Z",
         "endDateTime": "2099-01-11T00:00:00Z"
      },
      "version": "1.0",
      "specCharacteristic": [
         {
```

© TM Forum 2024. All Rights Reserved.

```
            "@type": "CharacteristicSpecification",
            "name": "identifier",
            "valueType": "string",
            "description": "Unique identifier for the API. This value will be used
as the audience parameter on authorization calls.",
            "configurable": true,
            "minCardinality": 1,
            "maxCardinality": 1
         },
         {
            "@type": "CharacteristicSpecification",
            "name": "maximumAccessTokenLifetime",
            "valueType": "integer",
            "description": "Time until an access token issued for this API will
expire.",
            "configurable": true,
            "minCardinality": 1,
            "maxCardinality": 1
         },
         {
            "@type": "CharacteristicSpecification",
            "name": "jwtTokenSigningAlgorithm",
            "valueType": "string",
            "description": "Algorithm used to sign access tokens issued for this
API.",
            "configurable": true,
            "minCardinality": 1,
            "maxCardinality": 1,
            "characteristicValueSpecification": [
               {
                  "@type": "StringCharacteristicValueSpecification",
                  "@baseType": "characteristicValueSpecification",
                  "isDefault": true,
                  "valueType": "string",
                  "value": "RS256 (RSA Signature with SHA-256)"
               },
               {
                  "@type": "StringCharacteristicValueSpecification",
                  "@baseType": "characteristicValueSpecification",
                  "valueType": "string",
                  "value": "HS256 (HMAC with SHA-256)"
               },
               {
                  "@type": "StringCharacteristicValueSpecification",
                  "@baseType": "characteristicValueSpecification",
                  "valueType": "string",
                  "value": "PS256 (RSA Signature with SHA-256)"
               }
            ]
         }
      ],
      "permissionSpecificationSet": [
         {
            "href": "https://serverRoot/tmf-
api/rolesAndPermissionsManagement/v5/permissionSpecificationSet/193c8d6887394ca5b502
90685db512ca",
            "id": "193c8d6887394ca5b50290685db512ca",
            "@type": "PermissionSpecificationSetRef"
         }
      ],
      "creationDate": "2024-02-20T17:00:00Z"
   }
```


© TM Forum 2024. All Rights Reserved.

```
]
```

Creates a ResourceRoleSpecification

POST /resourceRoleSpecification?fields=…

Description

This operation creates a ResourceRoleSpecification entity.

Mandatory Attributes

Mandatory Attributes	Rule
id	
lifecycleStatus	
name	
validFor	
@type	

Usage samples

Create Resource Role Specification example.

Request

```
POST /resourceRoleSpecification
Content-Type: application/json

{
   "@type": "ResourceRoleSpecification",
   "name": "TMF620 API provider role specification",
   "lifecycleStatus": "active",
   "validFor": {
      "startDateTime": "2024-02-20T17:00:00Z",
      "endDateTime": "2099-01-11T00:00:00Z"
   },
   "version": "1.0",
   "specCharacteristic": [
      {
         "@type": "CharacteristicSpecification",
         "name": "identifier",
         "valueType": "string",
         "description": "Unique identifier for the API. This value will be used as
the audience parameter on authorization calls.",
         "configurable": true,
         "minCardinality": 1,
         "maxCardinality": 1
      },
      {
         "@type": "CharacteristicSpecification",
         "name": "maximumAccessTokenLifetime",
         "valueType": "integer",
         "description": "Time until an access token issued for this API will
```

© TM Forum 2024. All Rights Reserved.

```
expire.",
         "configurable": true,
         "minCardinality": 1,
         "maxCardinality": 1
      },
      {
         "@type": "CharacteristicSpecification",
         "name": "jwtTokenSigningAlgorithm",
         "valueType": "string",
         "description": "Algorithm used to sign access tokens issued for this API.",
         "configurable": true,
         "minCardinality": 1,
         "maxCardinality": 1,
         "characteristicValueSpecification": [
            {
               "@type": "StringCharacteristicValueSpecification",
               "@baseType": "characteristicValueSpecification",
               "isDefault": true,
               "valueType": "string",
               "value": "RS256 (RSA Signature with SHA-256)"
            },
            {
               "@type": "StringCharacteristicValueSpecification",
               "@baseType": "characteristicValueSpecification",
               "valueType": "string",
               "value": "HS256 (HMAC with SHA-256)"
            },
            {
               "@type": "StringCharacteristicValueSpecification",
               "@baseType": "characteristicValueSpecification",
               "valueType": "string",
               "value": "PS256 (RSA Signature with SHA-256)"
            }
         ]
      }
   ],
   "permissionSpecificationSet": [
      {
         "id": "193c8d6887394ca5b50290685db512ca",
         "@type": "PermissionSpecificationSetRef"
      }
   ]
}
```

Response

```
201

{
   "@type": "ResourceRoleSpecification",
   "href": "https://serverRoot/tmf-
api/resourceRoleManagement/v5/resourceRoleSpecification/2391ee4acbc142b1a4b74b4d4ab3
8152",
   "id": "2391ee4acbc142b1a4b74b4d4ab38152",
   "name": "TMF620 API provider role specification",
   "lastUpdate": "2024-02-20T17:00:00Z",
   "lifecycleStatus": "active",
   "validFor": {
      "startDateTime": "2024-02-20T17:00:00Z",
      "endDateTime": "2099-01-11T00:00:00Z"
```

© TM Forum 2024. All Rights Reserved.

```
    },
    "version": "1.0",
    "specCharacteristic": [
       {
          "@type": "CharacteristicSpecification",
          "name": "identifier",
          "valueType": "string",
          "description": "Unique identifier for the API. This value will be used as
the audience parameter on authorization calls.",
          "configurable": true,
          "minCardinality": 1,
          "maxCardinality": 1
       },
       {
          "@type": "CharacteristicSpecification",
          "name": "maximumAccessTokenLifetime",
          "valueType": "integer",
          "description": "Time until an access token issued for this API will
expire.",
          "configurable": true,
          "minCardinality": 1,
          "maxCardinality": 1
       },
       {
          "@type": "CharacteristicSpecification",
          "name": "jwtTokenSigningAlgorithm",
          "valueType": "string",
          "description": "Algorithm used to sign access tokens issued for this API.",
          "configurable": true,
          "minCardinality": 1,
          "maxCardinality": 1,
          "characteristicValueSpecification": [
             {
                "@type": "StringCharacteristicValueSpecification",
                "@baseType": "characteristicValueSpecification",
                "isDefault": true,
                "valueType": "string",
                "value": "RS256 (RSA Signature with SHA-256)"
             },
             {
                "@type": "StringCharacteristicValueSpecification",
                "@baseType": "characteristicValueSpecification",
                "valueType": "string",
                "value": "HS256 (HMAC with SHA-256)"
             },
             {
                "@type": "StringCharacteristicValueSpecification",
                "@baseType": "characteristicValueSpecification",
                "valueType": "string",
                "value": "PS256 (RSA Signature with SHA-256)"
             }
          ]
       }
    ],
    "permissionSpecificationSet": [
       {
          "href": "https://serverRoot/tmf-
api/rolesAndPermissionsManagement/v5/permissionSpecificationSet/193c8d6887394ca5b502
90685db512ca",
          "id": "193c8d6887394ca5b50290685db512ca",
          "@type": "PermissionSpecificationSetRef"
       }
    ],
```


© TM Forum 2024. All Rights Reserved.

```
    "creationDate": "2024-02-20T17:00:00Z"
}
```

Updates partially a ResourceRoleSpecification

PATCH /resourceRoleSpecification/{id}?fields=…

Description

This operation allows partial updates of a ResourceRoleSpecification entity. Support of json/merge (https://tools.ietf.org/html/rfc7396) is mandatory, support of json/patch (http://tools.ietf.org/html/rfc5789) is optional. Note: If the update operation yields to the creation of sub-resources or relationships, the same rules concerning mandatory sub-resource attributes and default value settings in the POST operation applies to the PATCH operation. Hence these tables are not repeated here.

Patchable and Non Patchable Attributes

Non Patchable Attributes	Rule
creationDate	
href	
id	
@baseType	@baseType is immutable
@schemaLocation	@schemaLocation is immutable
@type	@type is immutable

Patchable Attributes	Rule
attachment	
constraint	
description	
entitySpecRelationship	
isBundle	
lastUpdate	
lifecycleStatus	
name	
permissionSpecificationSet	
relatedParty	
specCharacteristic	
targetEntitySchema	
validFor	
version	

© TM Forum 2024. All Rights Reserved.

Usage samples

PartialUpdate_ResourceRoleSpecification_using_json example.

Request

```
PATCH /resourceRoleSpecification/2391ee4acbc142b1a4b74b4d4ab38152
Content-Type: application/json

{
   "@type": "ResourceRoleSpecification",
   "name": "TMF620 API provider role specification",
   "permissionSpecificationSet": [
      {
         "id": "193c8d6887394ca5b50290685db512ca",
         "@type": "PermissionSpecificationSetRef"
      }
   ]
}
```

Response

```
200

{
   "@type": "ResourceRoleSpecification",
   "href": "https://serverRoot/tmf-
api/resourceRoleManagement/v5/resourceRoleSpecification/2391ee4acbc142b1a4b74b4d4ab3
8152",
   "id": "2391ee4acbc142b1a4b74b4d4ab38152",
   "name": "TMF620 API provider role specification",
   "lastUpdate": "2024-02-20T17:00:00Z",
   "lifecycleStatus": "active",
   "validFor": {
      "startDateTime": "2024-02-20T17:00:00Z",
      "endDateTime": "2099-01-11T00:00:00Z"
   },
   "version": "1.0",
   "specCharacteristic": [
      {
         "@type": "CharacteristicSpecification",
         "name": "identifier",
         "valueType": "string",
         "description": "Unique identifier for the API. This value will be used as
the audience parameter on authorization calls.",
         "configurable": true,
         "minCardinality": 1,
         "maxCardinality": 1
      },
      {
         "@type": "CharacteristicSpecification",
         "name": "maximumAccessTokenLifetime",
         "valueType": "integer",
         "description": "Time until an access token issued for this API will
expire.",
         "configurable": true,
         "minCardinality": 1,
         "maxCardinality": 1
```

© TM Forum 2024. All Rights Reserved.

```
      },
      {
         "@type": "CharacteristicSpecification",
         "name": "jwtTokenSigningAlgorithm",
         "valueType": "string",
         "description": "Algorithm used to sign access tokens issued for this API.",
         "configurable": true,
         "minCardinality": 1,
         "maxCardinality": 1,
         "characteristicValueSpecification": [
            {
               "@type": "StringCharacteristicValueSpecification",
               "@baseType": "characteristicValueSpecification",
               "isDefault": true,
               "valueType": "string",
               "value": "RS256 (RSA Signature with SHA-256)"
            },
            {
               "@type": "StringCharacteristicValueSpecification",
               "@baseType": "characteristicValueSpecification",
               "valueType": "string",
               "value": "HS256 (HMAC with SHA-256)"
            },
            {
               "@type": "StringCharacteristicValueSpecification",
               "@baseType": "characteristicValueSpecification",
               "valueType": "string",
               "value": "PS256 (RSA Signature with SHA-256)"
            }
         ]
      }
   ],
   "permissionSpecificationSet": [
      {
         "href": "https://serverRoot/tmf-
api/rolesAndPermissionsManagement/v5/permissionSpecificationSet/193c8d6887394ca5b502
90685db512ca",
         "id": "193c8d6887394ca5b50290685db512ca",
         "@type": "PermissionSpecificationSetRef"
      }
   ],
   "creationDate": "2024-02-20T17:00:00Z"
}
```


PartialUpdate_ResourceRoleSpecification_using_json_merge_patch example.

Request

```
PATCH /resourceRoleSpecification/2391ee4acbc142b1a4b74b4d4ab38152
Content-Type: application/merge-patch+json

{
   "@type": "ResourceRoleSpecification",
   "name": "TMF620 API provider role specification",
   "permissionSpecificationSet": [
      {
         "id": "193c8d6887394ca5b50290685db512ca",
         "@type": "PermissionSpecificationSetRef"
      }
   ]
```

© TM Forum 2024. All Rights Reserved.

```
}
```

Response

```
200

{
   "@type": "ResourceRoleSpecification",
   "href": "https://serverRoot/tmf-
api/resourceRoleManagement/v5/resourceRoleSpecification/2391ee4acbc142b1a4b74b4d4ab3
8152",
   "id": "2391ee4acbc142b1a4b74b4d4ab38152",
   "name": "TMF620 API provider role specification",
   "lastUpdate": "2024-02-20T17:00:00Z",
   "lifecycleStatus": "active",
   "validFor": {
      "startDateTime": "2024-02-20T17:00:00Z",
      "endDateTime": "2099-01-11T00:00:00Z"
   },
   "version": "1.0",
   "specCharacteristic": [
      {
         "@type": "CharacteristicSpecification",
         "name": "identifier",
         "valueType": "string",
         "description": "Unique identifier for the API. This value will be used as
the audience parameter on authorization calls.",
         "configurable": true,
         "minCardinality": 1,
         "maxCardinality": 1
      },
      {
         "@type": "CharacteristicSpecification",
         "name": "maximumAccessTokenLifetime",
         "valueType": "integer",
         "description": "Time until an access token issued for this API will
expire.",
         "configurable": true,
         "minCardinality": 1,
         "maxCardinality": 1
      },
      {
         "@type": "CharacteristicSpecification",
         "name": "jwtTokenSigningAlgorithm",
         "valueType": "string",
         "description": "Algorithm used to sign access tokens issued for this API.",
         "configurable": true,
         "minCardinality": 1,
         "maxCardinality": 1,
         "characteristicValueSpecification": [
            {
               "@type": "StringCharacteristicValueSpecification",
               "@baseType": "characteristicValueSpecification",
               "isDefault": true,
               "valueType": "string",
               "value": "RS256 (RSA Signature with SHA-256)"
            },
            {
               "@type": "StringCharacteristicValueSpecification",
               "@baseType": "characteristicValueSpecification",
```

© TM Forum 2024. All Rights Reserved.

```
                "valueType": "string",
                "value": "HS256 (HMAC with SHA-256)"
             },
             {
                "@type": "StringCharacteristicValueSpecification",
                "@baseType": "characteristicValueSpecification",
                "valueType": "string",
                "value": "PS256 (RSA Signature with SHA-256)"
             }
          ]
       }
    ],
    "permissionSpecificationSet": [
       {
          "href": "https://serverRoot/tmf-
api/rolesAndPermissionsManagement/v5/permissionSpecificationSet/193c8d6887394ca5b502
90685db512ca",
          "id": "193c8d6887394ca5b50290685db512ca",
          "@type": "PermissionSpecificationSetRef"
       }
    ],
    "creationDate": "2024-02-20T17:00:00Z"
}
```


PartialUpdate_ResourceRoleSpecification_using_json_patch example.

Request

```
PATCH /resourceRoleSpecification/2391ee4acbc142b1a4b74b4d4ab38152
Content-Type: application/json-patch+json

[
  {
    "op": "replace",
    "path": "/name",
    "value": "TMF620 API provider role specification"
  },
  {
    "op": "replace",
    "path": "/lifecycleStatus",
    "value": "active"
  }
]
```

Response


```
200

{
   "@type": "ResourceRoleSpecification",
   "href": "https://serverRoot/tmf-
api/resourceRoleManagement/v5/resourceRoleSpecification/2391ee4acbc142b1a4b74b4d4ab3
8152",
   "id": "2391ee4acbc142b1a4b74b4d4ab38152",
   "name": "TMF620 API provider role specification",
   "lastUpdate": "2024-02-20T17:00:00Z",
   "lifecycleStatus": "active",
```


© TM Forum 2024. All Rights Reserved.

```
   "validFor": {
      "startDateTime": "2024-02-20T17:00:00Z",
      "endDateTime": "2099-01-11T00:00:00Z"
   },
   "version": "1.0",
   "specCharacteristic": [
      {
         "@type": "CharacteristicSpecification",
         "name": "identifier",
         "valueType": "string",
         "description": "Unique identifier for the API. This value will be used as
the audience parameter on authorization calls.",
         "configurable": true,
         "minCardinality": 1,
         "maxCardinality": 1
      },
      {
         "@type": "CharacteristicSpecification",
         "name": "maximumAccessTokenLifetime",
         "valueType": "integer",
         "description": "Time until an access token issued for this API will
expire.",
         "configurable": true,
         "minCardinality": 1,
         "maxCardinality": 1
      },
      {
         "@type": "CharacteristicSpecification",
         "name": "jwtTokenSigningAlgorithm",
         "valueType": "string",
         "description": "Algorithm used to sign access tokens issued for this API.",
         "configurable": true,
         "minCardinality": 1,
         "maxCardinality": 1,
         "characteristicValueSpecification": [
            {
               "@type": "StringCharacteristicValueSpecification",
               "@baseType": "characteristicValueSpecification",
               "isDefault": true,
               "valueType": "string",
               "value": "RS256 (RSA Signature with SHA-256)"
            },
            {
               "@type": "StringCharacteristicValueSpecification",
               "@baseType": "characteristicValueSpecification",
               "valueType": "string",
               "value": "HS256 (HMAC with SHA-256)"
            },
            {
               "@type": "StringCharacteristicValueSpecification",
               "@baseType": "characteristicValueSpecification",
               "valueType": "string",
               "value": "PS256 (RSA Signature with SHA-256)"
            }
         ]
      }
   ],
   "permissionSpecificationSet": [
      {
         "href": "https://serverRoot/tmf-
api/rolesAndPermissionsManagement/v5/permissionSpecificationSet/193c8d6887394ca5b502
90685db512ca",
         "id": "193c8d6887394ca5b50290685db512ca",
```


© TM Forum 2024. All Rights Reserved.

```
            "@type": "PermissionSpecificationSetRef"
         }
      ],
      "creationDate": "2024-02-20T17:00:00Z"
   }
```

PartialUpdate_ResourceRoleSpecification_using_json_patch_query example.

Request

```
PATCH /resourceRoleSpecification/2391ee4acbc142b1a4b74b4d4ab38152
Content-Type: application/json-patch-query+json

[
  {
    "op": "replace",
    "path": "/specCharacteristic?name=maximumAccessTokenLifetime",
    "value": {
      "@type": "CharacteristicSpecification",
      "name": "maximumAccessTokenLifetime",
      "valueType": "integer",
      "description": "Time until an access token issued for this API will expire.",
      "configurable": true,
      "minCardinality": 1,
      "maxCardinality": 1
    }
  }
]
```

Response

```
200

{
   "@type": "ResourceRoleSpecification",
   "href": "https://serverRoot/tmf-
api/resourceRoleManagement/v5/resourceRoleSpecification/2391ee4acbc142b1a4b74b4d4ab3
8152",
   "id": "2391ee4acbc142b1a4b74b4d4ab38152",
   "name": "TMF620 API provider role specification",
   "lastUpdate": "2024-02-20T17:00:00Z",
   "lifecycleStatus": "active",
   "validFor": {
      "startDateTime": "2024-02-20T17:00:00Z",
      "endDateTime": "2099-01-11T00:00:00Z"
   },
   "version": "1.0",
   "specCharacteristic": [
      {
         "@type": "CharacteristicSpecification",
         "name": "identifier",
         "valueType": "string",
         "description": "Unique identifier for the API. This value will be used as
the audience parameter on authorization calls.",
         "configurable": true,
         "minCardinality": 1,
         "maxCardinality": 1
```

© TM Forum 2024. All Rights Reserved.

```
      },
      {
         "@type": "CharacteristicSpecification",
         "name": "maximumAccessTokenLifetime",
         "valueType": "integer",
         "description": "Time until an access token issued for this API will
expire.",
         "configurable": true,
         "minCardinality": 1,
         "maxCardinality": 1
      },
      {
         "@type": "CharacteristicSpecification",
         "name": "jwtTokenSigningAlgorithm",
         "valueType": "string",
         "description": "Algorithm used to sign access tokens issued for this API.",
         "configurable": true,
         "minCardinality": 1,
         "maxCardinality": 1,
         "characteristicValueSpecification": [
            {
               "@type": "StringCharacteristicValueSpecification",
               "@baseType": "characteristicValueSpecification",
               "isDefault": true,
               "valueType": "string",
               "value": "RS256 (RSA Signature with SHA-256)"
            },
            {
               "@type": "StringCharacteristicValueSpecification",
               "@baseType": "characteristicValueSpecification",
               "valueType": "string",
               "value": "HS256 (HMAC with SHA-256)"
            },
            {
               "@type": "StringCharacteristicValueSpecification",
               "@baseType": "characteristicValueSpecification",
               "valueType": "string",
               "value": "PS256 (RSA Signature with SHA-256)"
            }
         ]
      }
   ],
   "permissionSpecificationSet": [
      {
         "href": "https://serverRoot/tmf-
api/rolesAndPermissionsManagement/v5/permissionSpecificationSet/193c8d6887394ca5b502
90685db512ca",
         "id": "193c8d6887394ca5b50290685db512ca",
         "@type": "PermissionSpecificationSetRef"
      }
   ],
   "creationDate": "2024-02-20T17:00:00Z"
}
```


Deletes a ResourceRoleSpecification

`DELETE /resourceRoleSpecification/{id}`

Description

© TM Forum 2024. All Rights Reserved.

This operation deletes a ResourceRoleSpecification entity.

Usage samples

Remove Resource Role Specification example.

Request

```
DELETE /resourceRoleSpecification/2391ee4acbc142b1a4b74b4d4ab38152
Content-Type: application/json
```

Response

```
204
```

Operations on ResourceRole

Retrieves a ResourceRole by ID

`GET /resourceRole/{id}?fields=…`

Description

This operation retrieves a ResourceRole entity. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

Usage samples

Retrieve Resource Role example.

Request

```
GET /resourceRole/2391ee4acbc142b1a4b74b4d4ab38152
Content-Type: application/json
```

Response

```
200

{
   "@type": "ResourceRole",
   "href": "https://serverRoot/tmf-
api/resourceRoleManagement/v5/resourceRole/2391ee4acbc142b1a4b74b4d4ab38152",
   "id": "2391ee4acbc142b1a4b74b4d4ab38152",
   "name": "TMF620 API provider resource role",
   "lastUpdate": "2024-02-20T17:00:00Z",
   "lifecycleStatus": "active",
   "validFor": {
      "startDateTime": "2024-02-20T17:00:00Z",
```

© TM Forum 2024. All Rights Reserved.

```
      "endDateTime": "2099-01-11T00:00:00Z"
   },
   "version": "1.0",
   "role": "Api Provider",
   "entityRoleSpecification": {
      "href": "https://serverRoot/tmf-
api/resourceRoleManagement/v5/resourceRoleSpecification/2391ee4acbc142b1a4b74b4d4ab3
8152",
      "id": "2391ee4acbc142b1a4b74b4d4ab38152",
      "@type": "EntityRoleSpecificationRef",
      "@referredType": "ResourceRoleSpecification"
   },
   "characteristic": [
      {
         "@type": "StringCharacteristic",
         "@baseType": "Characteristic",
         "name": "identitier",
         "valueType": "string",
         "value": "https://serverRoot/tmf-api/productCatalogManagement/v5"
      },
      {
         "@type": "IntegerCharacteristic",
         "@baseType": "Characteristic",
         "name": "maximumAccessTokenLifetime",
         "valueType": "integer",
         "value": 86400
      },
      {
         "@type": "StringCharacteristic",
         "@baseType": "Characteristic",
         "name": "jwtSigningAlgorithm",
         "valueType": "string",
         "value": "RS256 (RSA Signature with SHA-256)"
      }
   ],
   "creationDate": "2024-02-20T17:00:00Z",
   "engagedResource": {
      "href": "https://serverRoot/tmf-
api/resourceManagement/v5/resource/e3479404feb943d2af7af92ffe4ec2b0",
      "id": "e3479404feb943d2af7af92ffe4ec2b0",
      "@type": "ResourceRef",
      "@referredType": "Resource"
   }
}
```


List or find ResourceRole objects

`GET /resourceRole?fields=…`

Description

This operation list ResourceRole entities. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

Usage samples

Retrieve Resource Roles example.

© TM Forum 2024. All Rights Reserved.

Request

```
GET /resourceRole
Content-Type: application/json
```

Response

```
200

[
   {
      "@type": "ResourceRole",
      "href": "https://serverRoot/tmf-
api/resourceRoleManagement/v5/resourceRole/2391ee4acbc142b1a4b74b4d4ab38152",
      "id": "2391ee4acbc142b1a4b74b4d4ab38152",
      "name": "TMF620 API provider resource role",
      "lastUpdate": "2024-02-20T17:00:00Z",
      "lifecycleStatus": "active",
      "validFor": {
         "startDateTime": "2024-02-20T17:00:00Z",
         "endDateTime": "2099-01-11T00:00:00Z"
      },
      "version": "1.0",
      "role": "Api Provider",
      "entityRoleSpecification": {
         "href": "https://serverRoot/tmf-
api/resourceRoleManagement/v5/resourceRoleSpecification/2391ee4acbc142b1a4b74b4d4ab3
8152",
         "id": "2391ee4acbc142b1a4b74b4d4ab38152",
         "@type": "EntityRoleSpecificationRef",
         "@referredType": "ResourceRoleSpecification"
      },
      "characteristic": [
         {
            "@type": "StringCharacteristic",
            "@baseType": "Characteristic",
            "name": "identitier",
            "valueType": "string",
            "value": "https://serverRoot/tmf-api/productCatalogManagement/v5"
         },
         {
            "@type": "IntegerCharacteristic",
            "@baseType": "Characteristic",
            "name": "maximumAccessTokenLifetime",
            "valueType": "integer",
            "value": 86400
         },
         {
            "@type": "StringCharacteristic",
            "@baseType": "Characteristic",
            "name": "jwtSigningAlgorithm",
            "valueType": "string",
            "value": "RS256 (RSA Signature with SHA-256)"
         }
      ],
      "creationDate": "2024-02-20T17:00:00Z",
      "engagedResource": {
         "href": "https://serverRoot/tmf-
api/resourceManagement/v5/resource/e3479404feb943d2af7af92ffe4ec2b0",
```

© TM Forum 2024. All Rights Reserved.

```
            "id": "e3479404feb943d2af7af92ffe4ec2b0",
            "@type": "ResourceRef",
            "@referredType": "Resource"
         }
      }
]
```

Creates a ResourceRole

POST /resourceRole?fields=…

Description

This operation creates a ResourceRole entity.

Mandatory Attributes

Mandatory Attributes	Rule
engagedResource	
engagedResource.@type	
engagedResource.id	
id	
lifecycleStatus	
name	
validFor	
@type	

Usage samples

Create Resource Role example.

Request

```
POST /resourceRole
Content-Type: application/json

{
   "@type": "ResourceRole",
   "name": "TMF620 API provider resource role",
   "lastUpdate": "2024-02-20T17:00:00Z",
   "lifecycleStatus": "active",
   "validFor": {
      "startDateTime": "2024-02-20T17:00:00Z",
      "endDateTime": "2099-01-11T00:00:00Z"
   },
   "version": "1.0",
   "role": "Api Provider",
   "entityRoleSpecification": {
      "id": "2391ee4acbc142b1a4b74b4d4ab38152",
      "@type": "EntityRoleSpecificationRef",
      "@referredType": "ResourceRoleSpecification"
```

© TM Forum 2024. All Rights Reserved.

```
    },
    "characteristic": [
       {
          "@type": "StringCharacteristic",
          "@baseType": "Characteristic",
          "name": "identitier",
          "valueType": "string",
          "value": "https://serverRoot/tmf-api/productCatalogManagement/v5"
       },
       {
          "@type": "IntegerCharacteristic",
          "@baseType": "Characteristic",
          "name": "maximumAccessTokenLifetime",
          "valueType": "integer",
          "value": 86400
       },
       {
          "@type": "StringCharacteristic",
          "@baseType": "Characteristic",
          "name": "jwtSigningAlgorithm",
          "valueType": "string",
          "value": "RS256 (RSA Signature with SHA-256)"
       }
    ],
    "engagedResource": {
       "id": "e3479404feb943d2af7af92ffe4ec2b0",
       "@type": "ResourceRef",
       "@referredType": "Resource"
    }
}
```

Response

```
201

{
   "@type": "ResourceRole",
   "href": "https://serverRoot/tmf-
api/resourceRoleManagement/v5/resourceRole/2391ee4acbc142b1a4b74b4d4ab38152",
   "id": "2391ee4acbc142b1a4b74b4d4ab38152",
   "name": "TMF620 API provider resource role",
   "lastUpdate": "2024-02-20T17:00:00Z",
   "lifecycleStatus": "active",
   "validFor": {
      "startDateTime": "2024-02-20T17:00:00Z",
      "endDateTime": "2099-01-11T00:00:00Z"
   },
   "version": "1.0",
   "role": "Api Provider",
   "entityRoleSpecification": {
      "href": "https://serverRoot/tmf-
api/resourceRoleManagement/v5/resourceRoleSpecification/2391ee4acbc142b1a4b74b4d4ab3
8152",
      "id": "2391ee4acbc142b1a4b74b4d4ab38152",
      "@type": "EntityRoleSpecificationRef",
      "@referredType": "ResourceRoleSpecification"
   },
   "characteristic": [
      {
         "@type": "StringCharacteristic",
```

© TM Forum 2024. All Rights Reserved.

```
            "@baseType": "Characteristic",
            "name": "identitier",
            "valueType": "string",
            "value": "https://serverRoot/tmf-api/productCatalogManagement/v5"
         },
         {
            "@type": "IntegerCharacteristic",
            "@baseType": "Characteristic",
            "name": "maximumAccessTokenLifetime",
            "valueType": "integer",
            "value": 86400
         },
         {
            "@type": "StringCharacteristic",
            "@baseType": "Characteristic",
            "name": "jwtSigningAlgorithm",
            "valueType": "string",
            "value": "RS256 (RSA Signature with SHA-256)"
         }
      ],
      "creationDate": "2024-02-20T17:00:00Z",
      "engagedResource": {
         "href": "https://serverRoot/tmf-
   api/resourceManagement/v5/resource/e3479404feb943d2af7af92ffe4ec2b0",
         "id": "e3479404feb943d2af7af92ffe4ec2b0",
         "@type": "ResourceRef",
         "@referredType": "Resource"
      }
   }
```


Updates partially a ResourceRole

`PATCH /resourceRole/{id}?fields=…`

Description

This operation allows partial updates of a ResourceRole entity. Support of json/merge (https://tools.ietf.org/html/rfc7396) is mandatory, support of json/patch (http://tools.ietf.org/html/rfc5789) is optional. Note: If the update operation yields to the creation of sub-resources or relationships, the same rules concerning mandatory sub-resource attributes and default value settings in the POST operation applies to the PATCH operation. Hence these tables are not repeated here.

Patchable and Non Patchable Attributes

Non Patchable Attributes	Rule
creationDate	
href	
id	
@baseType	@baseType is immutable
@schemaLocation	@schemaLocation is immutable
@type	@type is immutable

© TM Forum 2024. All Rights Reserved.

Patchable Attributes	Rule
characteristic	
description	
engagedResource	
entityRoleSpecification	
isBundle	
lastUpdate	
lifecycleStatus	
name	
role	
validFor	
version	

Usage samples

PartialUpdate_ResourceRole_using_json example.

Request

```
PATCH /resourceRole/2391ee4acbc142b1a4b74b4d4ab38152
Content-Type: application/json

{
   "@type": "ResourceRole",
   "role": "Api Provider",
   "entityRoleSpecification": {
      "id": "2391ee4acbc142b1a4b74b4d4ab38152",
      "@type": "EntityRoleSpecificationRef",
      "@referredType": "ResourceRoleSpecification"
   }
}
```

Response

```
200

{
   "@type": "ResourceRole",
   "href": "https://serverRoot/tmf-
api/resourceRoleManagement/v5/resourceRole/2391ee4acbc142b1a4b74b4d4ab38152",
   "id": "2391ee4acbc142b1a4b74b4d4ab38152",
   "name": "TMF620 API provider resource role",
   "lastUpdate": "2024-02-20T17:00:00Z",
   "lifecycleStatus": "active",
   "validFor": {
      "startDateTime": "2024-02-20T17:00:00Z",
      "endDateTime": "2099-01-11T00:00:00Z"
   },
   "version": "1.0",
```

© TM Forum 2024. All Rights Reserved.

```
   "role": "Api Provider",
   "entityRoleSpecification": {
      "href": "https://serverRoot/tmf-
api/resourceRoleManagement/v5/resourceRoleSpecification/2391ee4acbc142b1a4b74b4d4ab3
8152",
      "id": "2391ee4acbc142b1a4b74b4d4ab38152",
      "@type": "EntityRoleSpecificationRef",
      "@referredType": "ResourceRoleSpecification"
   },
   "characteristic": [
      {
         "@type": "StringCharacteristic",
         "@baseType": "Characteristic",
         "name": "identitier",
         "valueType": "string",
         "value": "https://serverRoot/tmf-api/productCatalogManagement/v5"
      },
      {
         "@type": "IntegerCharacteristic",
         "@baseType": "Characteristic",
         "name": "maximumAccessTokenLifetime",
         "valueType": "integer",
         "value": 86400
      },
      {
         "@type": "StringCharacteristic",
         "@baseType": "Characteristic",
         "name": "jwtSigningAlgorithm",
         "valueType": "string",
         "value": "RS256 (RSA Signature with SHA-256)"
      }
   ],
   "creationDate": "2024-02-20T17:00:00Z",
   "engagedResource": {
      "href": "https://serverRoot/tmf-
api/resourceManagement/v5/resource/e3479404feb943d2af7af92ffe4ec2b0",
      "id": "e3479404feb943d2af7af92ffe4ec2b0",
      "@type": "ResourceRef",
      "@referredType": "Resource"
   }
}
```


PartialUpdate_ResourceRole_using_json_merge_patch

Request

```
PATCH /resourceRole/2391ee4acbc142b1a4b74b4d4ab38152
Content-Type: application/merge-patch+json

{
   "@type": "ResourceRole",
   "role": "Api Provider",
   "entityRoleSpecification": {
      "id": "2391ee4acbc142b1a4b74b4d4ab38152",
      "@type": "EntityRoleSpecificationRef",
      "@referredType": "ResourceRoleSpecification"
   }
}
```

© TM Forum 2024. All Rights Reserved.

Response


```
200

{
   "@type": "ResourceRole",
   "href": "https://serverRoot/tmf-
api/resourceRoleManagement/v5/resourceRole/2391ee4acbc142b1a4b74b4d4ab38152",
   "id": "2391ee4acbc142b1a4b74b4d4ab38152",
   "name": "TMF620 API provider resource role",
   "lastUpdate": "2024-02-20T17:00:00Z",
   "lifecycleStatus": "active",
   "validFor": {
      "startDateTime": "2024-02-20T17:00:00Z",
      "endDateTime": "2099-01-11T00:00:00Z"
   },
   "version": "1.0",
   "role": "Api Provider",
   "entityRoleSpecification": {
      "href": "https://serverRoot/tmf-
api/resourceRoleManagement/v5/resourceRoleSpecification/2391ee4acbc142b1a4b74b4d4ab3
8152",
      "id": "2391ee4acbc142b1a4b74b4d4ab38152",
      "@type": "EntityRoleSpecificationRef",
      "@referredType": "ResourceRoleSpecification"
   },
   "characteristic": [
      {
         "@type": "StringCharacteristic",
         "@baseType": "Characteristic",
         "name": "identitier",
         "valueType": "string",
         "value": "https://serverRoot/tmf-api/productCatalogManagement/v5"
      },
      {
         "@type": "IntegerCharacteristic",
         "@baseType": "Characteristic",
         "name": "maximumAccessTokenLifetime",
         "valueType": "integer",
         "value": 86400
      },
      {
         "@type": "StringCharacteristic",
         "@baseType": "Characteristic",
         "name": "jwtSigningAlgorithm",
         "valueType": "string",
         "value": "RS256 (RSA Signature with SHA-256)"
      }
   ],
   "creationDate": "2024-02-20T17:00:00Z",
   "engagedResource": {
      "href": "https://serverRoot/tmf-
api/resourceManagement/v5/resource/e3479404feb943d2af7af92ffe4ec2b0",
      "id": "e3479404feb943d2af7af92ffe4ec2b0",
      "@type": "ResourceRef",
      "@referredType": "Resource"
   }
}
```


PartialUpdate_ResourceRole_using_json_patch example.

© TM Forum 2024. All Rights Reserved.

Request

```
PATCH /resourceRole/2391ee4acbc142b1a4b74b4d4ab38152
Content-Type: application/json-patch+json

[
  {
    "op": "replace",
    "path": "/name",
    "value": "TMF620 API provider resource role"
  },
  {
    "op": "replace",
    "path": "/role",
    "value": "Api Provider"
  }
]
```

Response

```
200

{
   "@type": "ResourceRole",
   "href": "https://serverRoot/tmf-
api/resourceRoleManagement/v5/resourceRole/2391ee4acbc142b1a4b74b4d4ab38152",
   "id": "2391ee4acbc142b1a4b74b4d4ab38152",
   "name": "TMF620 API provider resource role",
   "lastUpdate": "2024-02-20T17:00:00Z",
   "lifecycleStatus": "active",
   "validFor": {
      "startDateTime": "2024-02-20T17:00:00Z",
      "endDateTime": "2099-01-11T00:00:00Z"
   },
   "version": "1.0",
   "role": "Api Provider",
   "entityRoleSpecification": {
      "href": "https://serverRoot/tmf-
api/resourceRoleManagement/v5/resourceRoleSpecification/2391ee4acbc142b1a4b74b4d4ab3
8152",
      "id": "2391ee4acbc142b1a4b74b4d4ab38152",
      "@type": "EntityRoleSpecificationRef",
      "@referredType": "ResourceRoleSpecification"
   },
   "characteristic": [
      {
         "@type": "StringCharacteristic",
         "@baseType": "Characteristic",
         "name": "identitier",
         "valueType": "string",
         "value": "https://serverRoot/tmf-api/productCatalogManagement/v5"
      },
      {
         "@type": "IntegerCharacteristic",
         "@baseType": "Characteristic",
         "name": "maximumAccessTokenLifetime",
         "valueType": "integer",
         "value": 86400
      },
```

© TM Forum 2024. All Rights Reserved.

```
      {
         "@type": "StringCharacteristic",
         "@baseType": "Characteristic",
         "name": "jwtSigningAlgorithm",
         "valueType": "string",
         "value": "RS256 (RSA Signature with SHA-256)"
      }
   ],
   "creationDate": "2024-02-20T17:00:00Z",
   "engagedResource": {
      "href": "https://serverRoot/tmf-
api/resourceManagement/v5/resource/e3479404feb943d2af7af92ffe4ec2b0",
      "id": "e3479404feb943d2af7af92ffe4ec2b0",
      "@type": "ResourceRef",
      "@referredType": "Resource"
   }
}
```


PartialUpdate_ResourceRole_using_json_patch_query example.

Request

```
PATCH /resourceRole/2391ee4acbc142b1a4b74b4d4ab38152
Content-Type: application/json-patch-query+json

[
  {
    "op": "replace",
    "path": "/characteristic?name=maximumAccessTokenLifetime",
    "value": {
      "@type": "IntegerCharacteristic",
      "@baseType": "Characteristic",
      "name": "maximumAccessTokenLifetime",
      "valueType": "integer",
      "value": 86400
    }
  }
]
```

Response


```
200

{
   "@type": "ResourceRole",
   "href": "https://serverRoot/tmf-
api/resourceRoleManagement/v5/resourceRole/2391ee4acbc142b1a4b74b4d4ab38152",
   "id": "2391ee4acbc142b1a4b74b4d4ab38152",
   "name": "TMF620 API provider resource role",
   "lastUpdate": "2024-02-20T17:00:00Z",
   "lifecycleStatus": "active",
   "validFor": {
      "startDateTime": "2024-02-20T17:00:00Z",
      "endDateTime": "2099-01-11T00:00:00Z"
   },
   "version": "1.0",
   "role": "Api Provider",
```


© TM Forum 2024. All Rights Reserved.

```
    "entityRoleSpecification": {
       "href": "https://serverRoot/tmf-
api/resourceRoleManagement/v5/resourceRoleSpecification/2391ee4acbc142b1a4b74b4d4ab3
8152",
       "id": "2391ee4acbc142b1a4b74b4d4ab38152",
       "@type": "EntityRoleSpecificationRef",
       "@referredType": "ResourceRoleSpecification"
    },
    "characteristic": [
       {
          "@type": "StringCharacteristic",
          "@baseType": "Characteristic",
          "name": "identitier",
          "valueType": "string",
          "value": "https://serverRoot/tmf-api/productCatalogManagement/v5"
       },
       {
          "@type": "IntegerCharacteristic",
          "@baseType": "Characteristic",
          "name": "maximumAccessTokenLifetime",
          "valueType": "integer",
          "value": 86400
       },
       {
          "@type": "StringCharacteristic",
          "@baseType": "Characteristic",
          "name": "jwtSigningAlgorithm",
          "valueType": "string",
          "value": "RS256 (RSA Signature with SHA-256)"
       }
    ],
    "creationDate": "2024-02-20T17:00:00Z",
    "engagedResource": {
       "href": "https://serverRoot/tmf-
api/resourceManagement/v5/resource/e3479404feb943d2af7af92ffe4ec2b0",
       "id": "e3479404feb943d2af7af92ffe4ec2b0",
       "@type": "ResourceRef",
       "@referredType": "Resource"
    }
}
```


Deletes a ResourceRole

`DELETE /resourceRole/{id}`

Description

This operation deletes a ResourceRole entity.

Usage samples

Remove Resource Role example.

Request

```
DELETE /resourceRole/2391ee4acbc142b1a4b74b4d4ab38152
Content-Type: application/json
```

© TM Forum 2024. All Rights Reserved.

Response

```
204
```

© TM Forum 2024. All Rights Reserved.

API NOTIFICATIONS

For every single of operation on the entities use the following templates and provide sample REST notification POST calls.

It is assumed that the Pub/Sub uses the Register and UnRegister mechanisms described in the REST Guidelines reproduced below.

Register listener

POST /hub

Description

Sets the communication endpoint address the service instance must use to deliver information about its health state, execution state, failures and metrics. Subsequent POST calls will be rejected by the service if it does not support multiple listeners. In this case DELETE /api/hub/{id} must be called before an endpoint can be created again.

Behavior

Returns HTTP/1.1 status code 204 if the request was successful.

Returns HTTP/1.1 status code 409 if request is not successful.

Usage Samples

Here's an example of a request for registering a listener.

Request

```
POST /api/hub

Accept: application/json

{
    "callback": "http://in.listener.com"
}
```

Response

```
201

Content-Type: application/json

Location: /api/hub/42

{
    "id":"42",
    "callback": "http://in.listener.com",
    "query":""
}
```

© TM Forum 2024. All Rights Reserved.

Unregister listener

DELETE /hub/{id}

Description

Clears the communication endpoint address that was set by creating the Hub..

Behavior

Returns HTTP/1.1 status code 204 if the request was successful.

Returns HTTP/1.1 status code 404 if the resource is not found.

Usage Samples

Here's an example of a request for un-registering a listener.

Request

```
DELETE /api/hub/42

Accept: application/json
```

Response

```
204
```

Publish Event to listener

POST /client/listener

Description

Clears the communication endpoint address that was set by creating the Hub.

Provides to a registered listener the description of the event that was raised. The /client/listener url is the callback url passed when registering the listener.

Behavior

Returns HTTP/1.1 status code 201 if the service is able to set the configuration.

Usage Samples

Here's an example of a notification received by the listener. In this example "EVENT TYPE" should be replaced by one of the notification types supported by this API (see Notification resources Models section) and EVENT BODY refers to the data structure of the given notification type.

Request

© TM Forum 2024. All Rights Reserved.

```
POST /client/listener

Accept: application/json

{
    "event": {
        EVENT BODY
    },
    "eventType": "EVENT_TYPE"
}
```

Response

```
201
```

For detailed examples on the general TM Forum notification mechanism, see the TMF REST Design Guidelines.

© TM Forum 2024. All Rights Reserved.

Acknowledgements

Version History

Version Number	Date Modified	Modified by:	Description of changes
5.0.0	13-Mar-2024	Bruno Sousa Fernandes (NOS)	Initial revision of the document.
5.0.0	08-Oct-2024	Bruno Sousa Fernandes (NOS)	Fixed examples according to new API tooling linter.

Release History

Release Number	Release Status	Date	Release led by:	Description
5.0.0	Preview	14-Nov-2024	Bruno Sousa Fernandes (NOS)	Initial revision of the document.

Contributors to Document

This document was prepared by the members of the TM Forum API Project team:

Name	Company	Role
Bruno Sousa Fernandes, bruno.sfernandes@nos.pt	NOS, PT	Author
Jonathan Goldberg, jonathan.goldberg@amdocs.com	Amdocs, IL	Reviewer
Pierre Gauthier, pgauthier@tmforum.org	TM Forum	Reviewer

© TM Forum 2024. All Rights Reserved.